

22050 Continuous-Time Signals and Linear Systems

Kaj-Åge Henneberg

L11

Bode plot 2: Quadratic factors

Filter design by placement of poles and zeros

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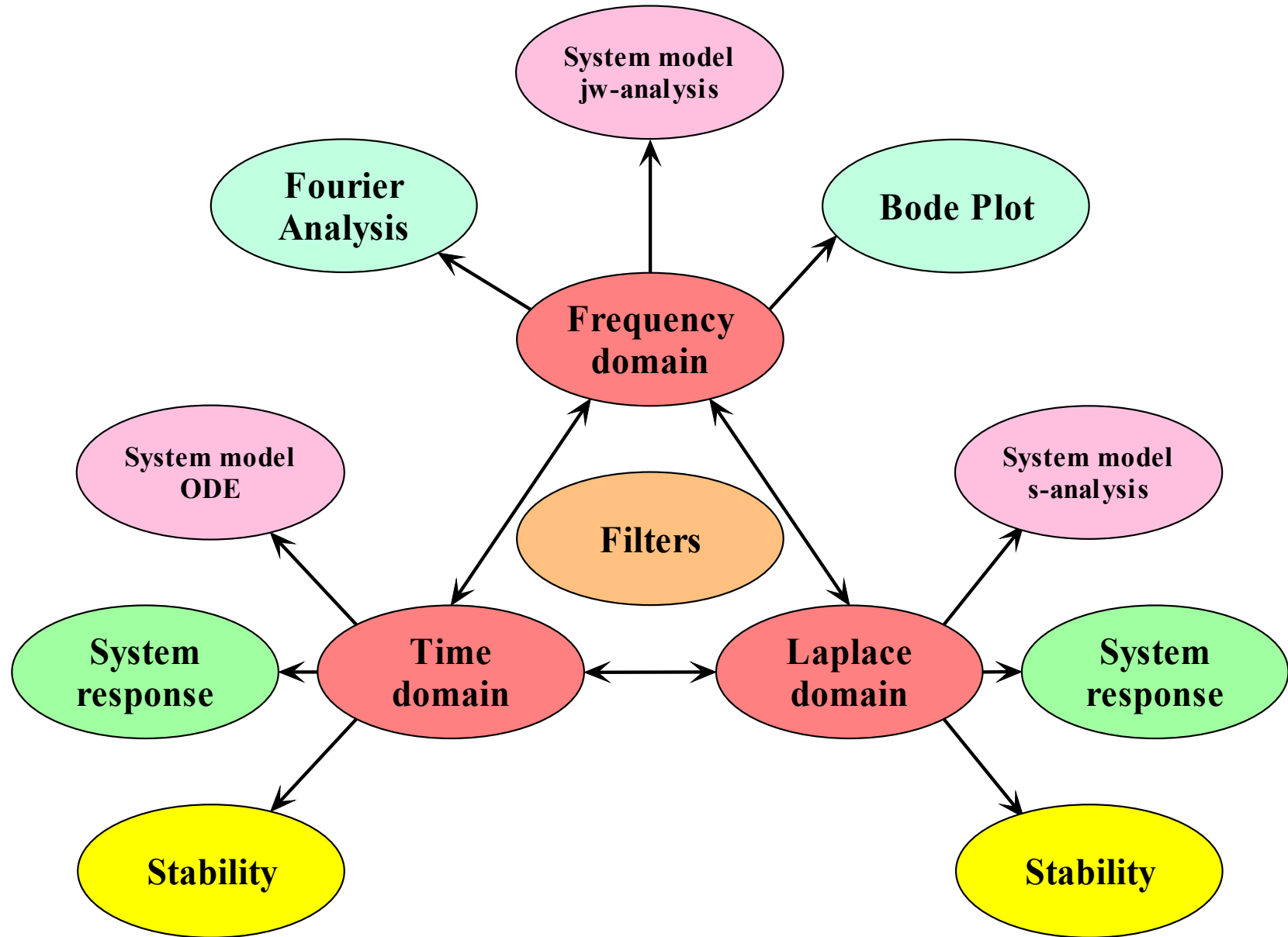
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7.2 Bode plot 2: Quadratic factors

7.4 Filter design by placement of poles and zeros

Bode plot 2

Constants and linear factors

Video

Examples of drawing Bode plots

Draw the amplitude and phase responses for:

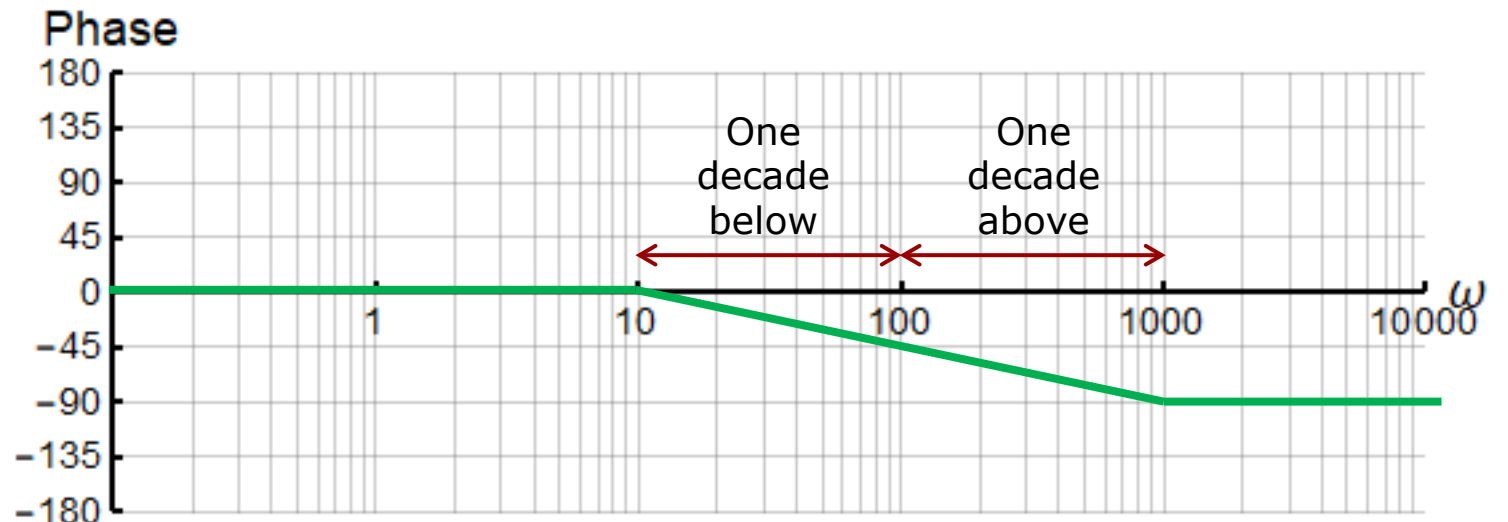
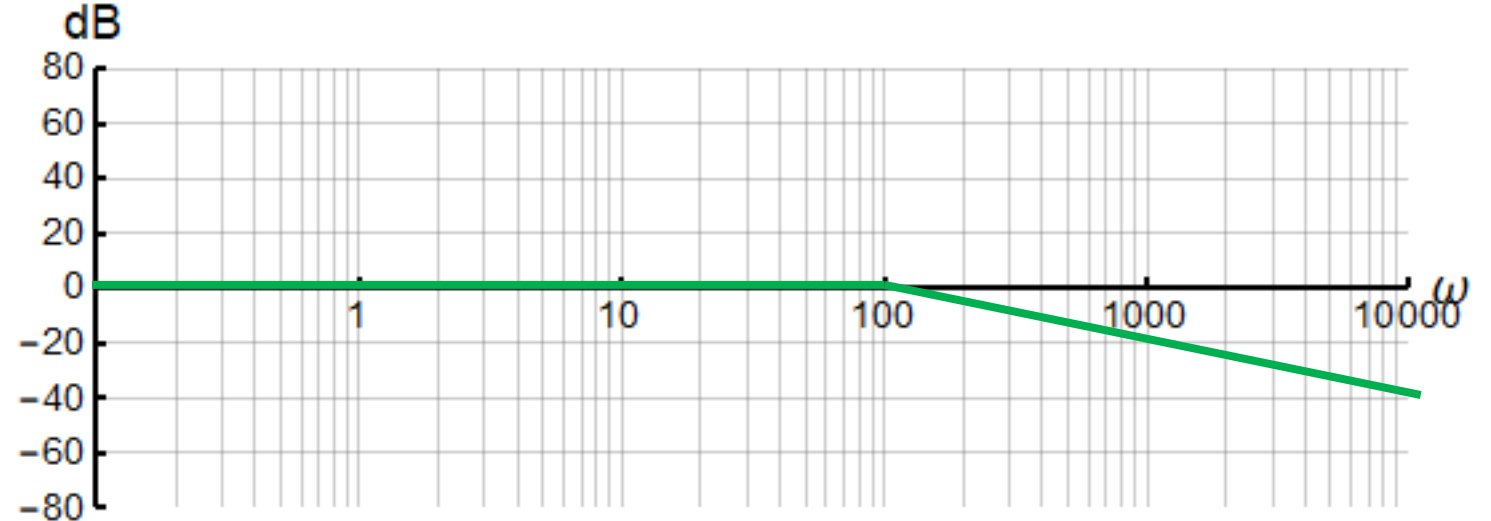
$$H(j\omega) = \frac{1}{\frac{j\omega}{100} + 1}$$

Gain:

DC gain: 1
Cut-off: 100 rad/s
Slope: -20 dB/decade

Phase:

DC phase: 0
HF phase: -90
Transition: 2 decades
 ω_L : 10 rad/s
 ω_H : 1000 rad/s



Examples of drawing Bode plots

Draw the amplitude and phase responses for:

$$H(j\omega) = \frac{j\omega}{\left(\frac{j\omega}{10} + 1\right)^2}$$

$$\frac{Me^{j\phi}}{Ne^{j\theta}} = \frac{M}{N} e^{j\phi} e^{-j\theta} = \frac{M}{N} e^{j(\phi-\theta)}$$

Gain:

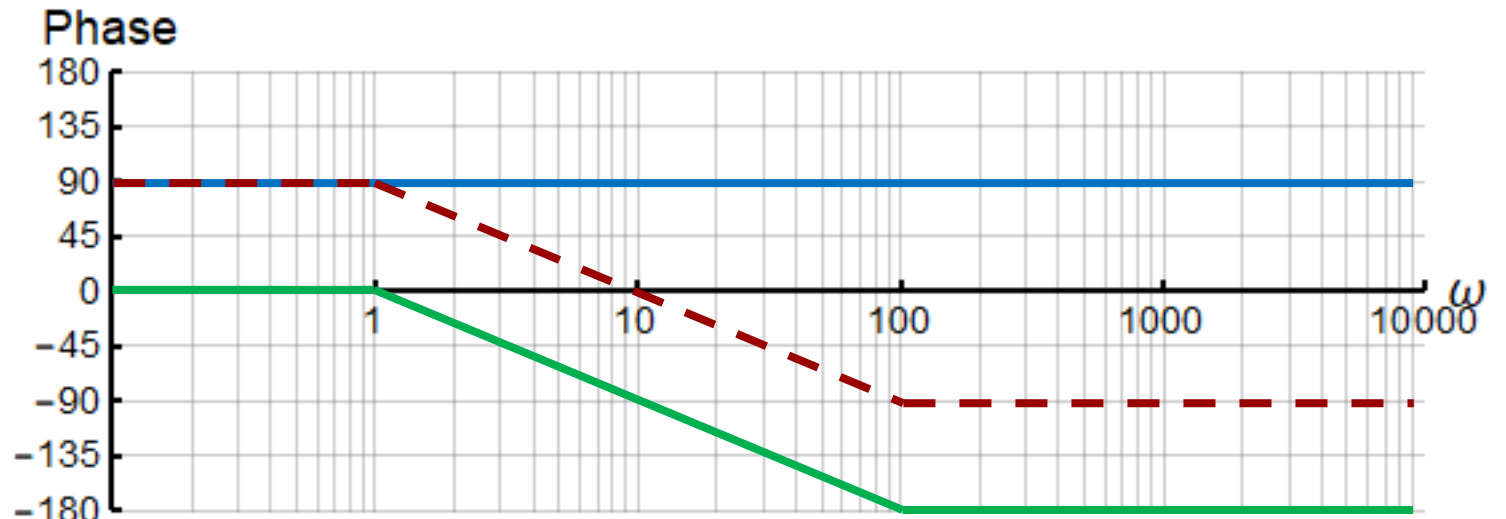
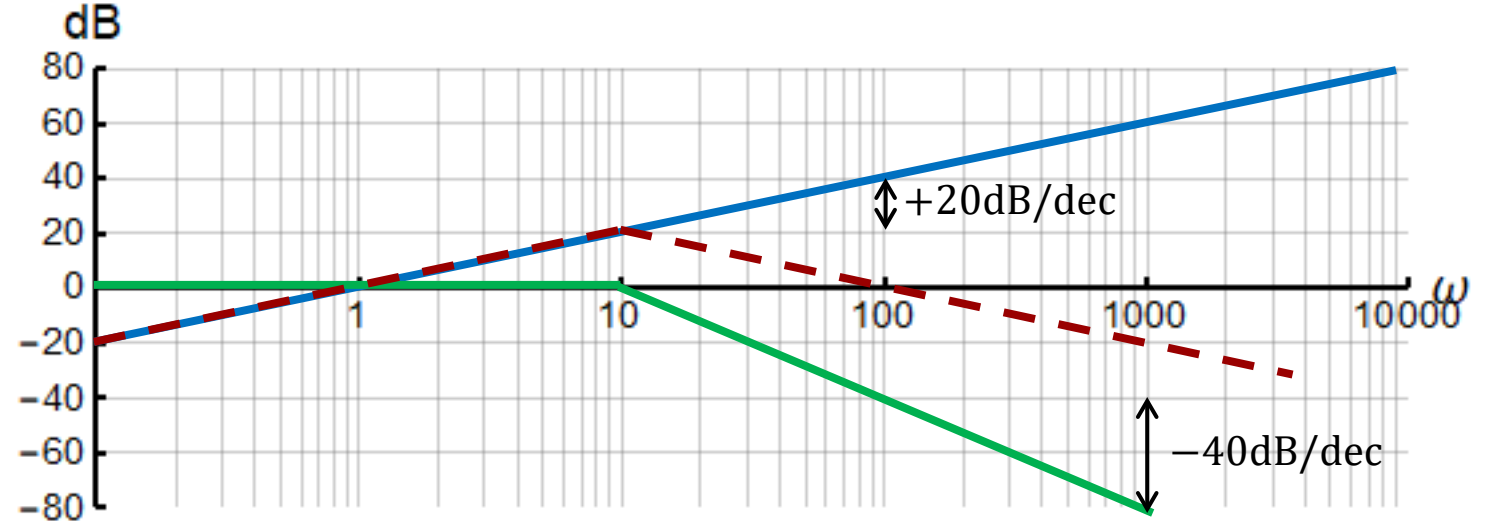
DC gain: 0
Cut-off: 10 rad/s
HF Slope: -40 dB/decade

Phase:

DC phase: 90
HF phase: -90
Transition: 2 decades

ω_L : 1 rad/s

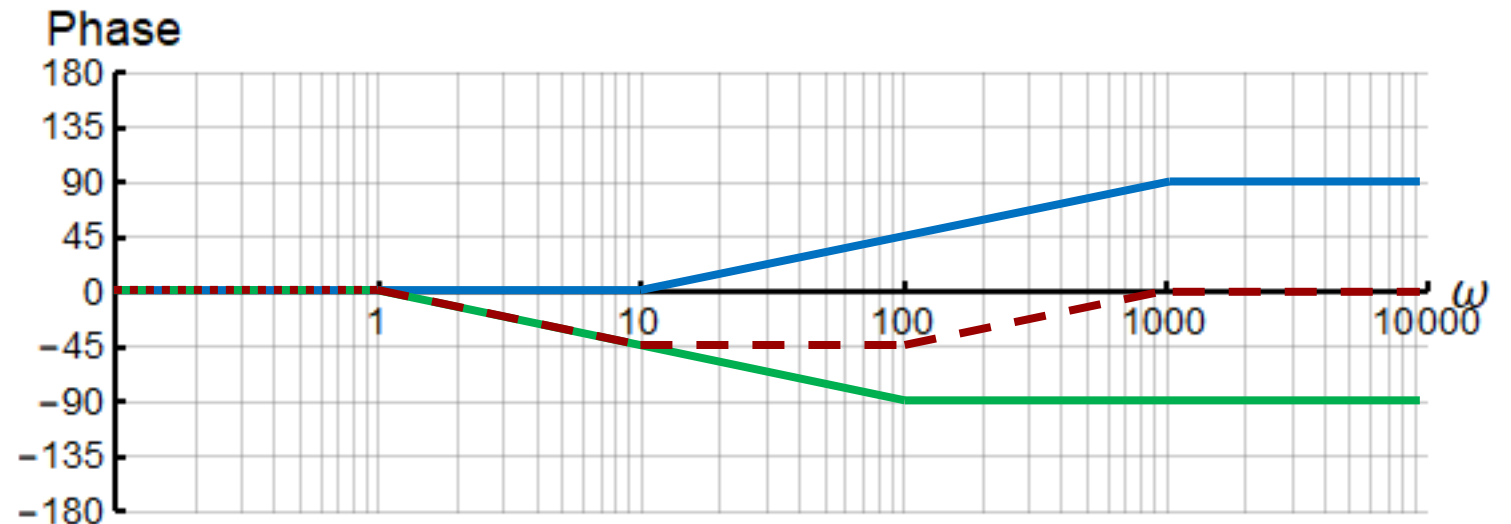
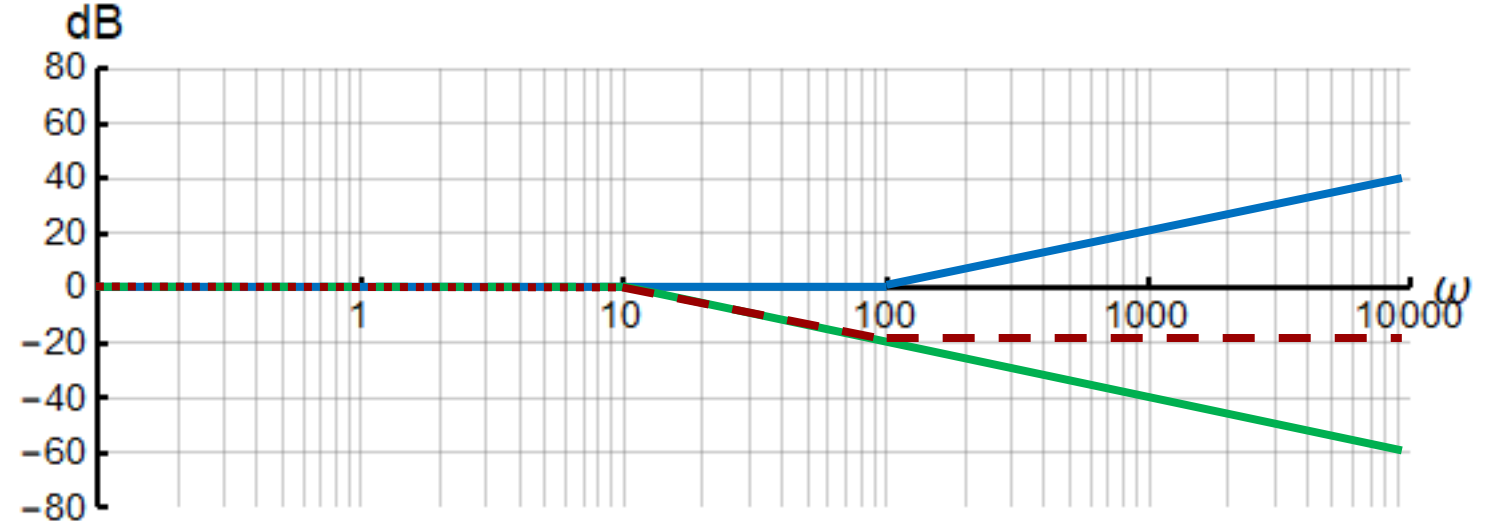
ω_H : 100 rad/s



Examples of drawing Bode plots

Draw the amplitude and phase responses for:

$$H(j\omega) = \frac{\frac{j\omega}{100} + 1}{\frac{j\omega}{10} + 1}$$



Bode plot 2

Quadratic factors

Video

Standard form for Bode plot

The advantage of taking $\log_{10}$ is that each factor now becomes a term. We can now draw an amplitude and phase curve for each factor, and then we can add the curves together graphically.

As each term may originate from a specific sub-system, we can easily identify the behavior of each sub-system and then correct those where needed.

Furthermore, if we decide to add a correcting subsystem into our system, we just add its Bode plot to the Bode plot of the original system.

$$\begin{aligned}
 |H(j\omega)|_{dB} = & \underbrace{20 \log_{10} K \frac{b_1 b_2^{m_2} b_4}{a_1 a_2^{n_2} a_4^{n_3}}}_{\text{constant}} + \underbrace{p \times 20 \log_{10} |j\omega|}_{\text{zero at origin}} + \underbrace{20 \log_{10} \left| \frac{j\omega}{b_1} + 1 \right|}_{\text{first order zero}} \\
 & + \underbrace{m_2 \times 20 \log_{10} \left| \frac{j\omega}{b_2} + 1 \right|}_{\text{repeated first order zero}} + \underbrace{m_3 \times 20 \log_{10} \left| \frac{(j\omega)^2}{b_4} + \frac{b_3}{b_4} j\omega + 1 \right|}_{\text{repeated second order zero}} \\
 & - \underbrace{q \times 20 \log_{10} |j\omega|}_{\text{pole at origin}} - \underbrace{20 \log_{10} \left| \frac{j\omega}{a_1} + 1 \right|}_{\text{first order pole}} - \underbrace{n_2 \times 20 \log_{10} \left| \frac{j\omega}{a_2} + 1 \right|}_{\text{repeated first order pole}} \\
 & - \underbrace{n_3 \times 20 \log_{10} \left| \frac{(j\omega)^2}{a_4} + \frac{a_3}{a_4} j\omega + 1 \right|}_{\text{repeated second order pole}}
 \end{aligned}$$

$$\begin{aligned}
 \angle H(\omega) = & p \angle(j\omega) + \angle \left(\frac{j\omega}{b_1} + 1 \right) + m_2 \angle \left(\frac{j\omega}{b_2} + 1 \right) + m_3 \angle \left(\frac{(j\omega)^2}{b_4} + \frac{b_3}{b_4} j\omega + 1 \right) \\
 & - q \angle(j\omega) - \angle \left(\frac{j\omega}{a_1} + 1 \right) - n_2 \angle \left(\frac{j\omega}{a_2} + 1 \right) - n_3 \angle \left(\frac{(j\omega)^2}{a_4} + \frac{a_3}{a_4} j\omega + 1 \right)
 \end{aligned}$$

When the transfer function contains a quadratic factor, we must first examine if it is a true quadratic factor, or if it can be factorized into one double linear factor (critically damped system) or two single linear factors (overdamped system).

We can do this easily by rewriting the quadratic factor from its mathematical form to its physical form:

Undamped
oscillation
frequency:

$$\omega_n = \sqrt{a_0}$$

Damping
factor:

$$\zeta = \frac{a_1}{2\sqrt{a_0}}$$

$$H(s) = \frac{b_0}{s^2 + a_1s + a_0} = \frac{b_0}{a_0} \frac{\omega_n^2}{s^2 + 2\zeta\omega_ns + \omega_n^2}$$

The quadratic factor should be kept as a quadratic factor if:

$$0 < \zeta < 1$$

Otherwise, the quadratic factor should be factorized.

We can write the amplitude of quadratic factors as:

$$|Q(j\omega)|_{dB} = m_3 \cdot 20 \log \left| \left(\frac{j\omega}{\omega_n} \right)^2 + 2j\zeta \left(\frac{\omega}{\omega_n} \right) + 1 \right|$$

To simplify the drawing of the quadratic function amplitude we consider two cases:

Low frequency

asymptote:

$$\omega \ll \omega_n$$

$$|Q(j\omega)|_{dB} = m_3 \cdot 20 \log |1| = 0 \text{ dB}$$

High frequency

asymptote:

$$\omega \gg \omega_n$$

$$|Q(j\omega)|_{dB} = m_3 \cdot 20 \log \left| \left(-\frac{\omega}{\omega_n} \right)^2 \right| = m_3 \cdot 40 \cdot \log \left| \left(\frac{\omega}{\omega_n} \right) \right|$$

Break

frequency:

$$\omega_{3dB} = \omega_n \sqrt{1 - 2\zeta^2 + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}$$

The deviation of the actual amplitude curve from the straight line asymptotes depends on the damping factor, thus we need more information to sketch a good approximation to the amplitude response.

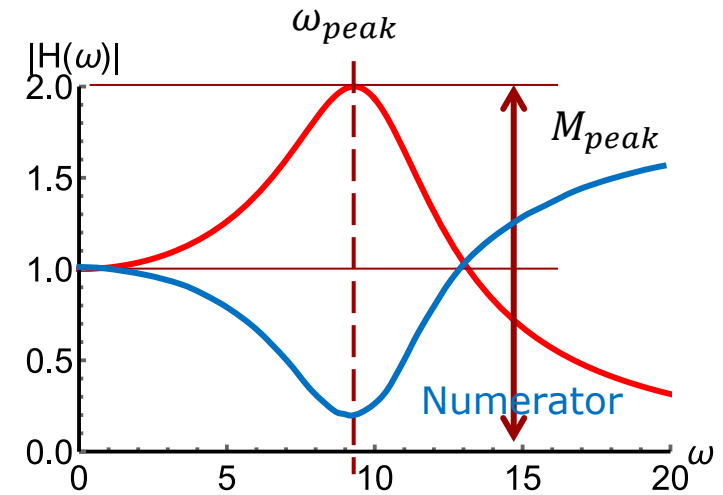
Quadratic Factor Amplitude Peak

The quadratic factor amplitude features a local maximum located at the frequency:

$$\omega_{pk} = \omega_n \sqrt{1 - 2\zeta^2}$$

Typo in book

Page 666!



When: $0 < \zeta < 1/\sqrt{2}$

the amplitude of the peak is:

$$M_{pk,dB} = m_3 \cdot 20 \cdot \log \left[2\zeta \sqrt{1 - \zeta^2} \right] \text{ dB}$$

$$M_{peak} = \frac{1}{2\zeta \sqrt{1 - \zeta^2}}$$

When: $1/\sqrt{2} \leq \zeta < 1$

the local peak is located at: $\omega_{pk} = 0$

and the peak amplitude is: $M_{pk} = 0 \text{ dB}$

Resonance

frequency: $\omega = \omega_n$

At the resonance frequency, the amplitude is:

$$|Q(j\omega_n)|_{dB} = m_3 \cdot 20 \log |j2\zeta|$$

We can write the phase of quadratic factors as:

$$\angle Q(j\omega) = m_3 \cdot \angle \left(\left(\frac{j\omega}{\omega_n} \right)^2 + 2j\zeta \left(\frac{\omega}{\omega_n} \right) + 1 \right)$$

Again, we consider two cases:

Low frequency
asymptote:

$$\omega \ll \omega_n$$

$$\angle Q(j\omega) = m_3 \cdot \angle(j0 + 1) = 0^\circ$$

High frequency
asymptote:

$$\omega \gg \omega_n$$

$$\angle Q(j\omega) = m_3 \cdot \angle \left(\left(\frac{j\omega}{\omega_n} \right)^2 \right) = m_3 \cdot \angle \left(- \left(\frac{\omega}{\omega_n} \right)^2 \right) = m_3 \cdot 180^\circ$$

Resonance
frequency:

$$\omega = \omega_n$$

$$\angle Q(j\omega_n) = m_3 \cdot \angle(-1 + 2\zeta j + 1) = m_3 \cdot 90^\circ$$

Group delay for quadratic factor

We derive the **group delay** from the phase spectrum:

$$t_g = -\frac{d\theta(\omega)}{d\omega}$$

For a quadratic factor, the group delay is zero at infinite frequency (all capacitors are short circuited).

At the resonance frequency the group delay will feature a peak equal to the effective time constant of the step response envelope.

The group delay at DC is reduced by 2ζ relative to the peak delay.

$$\angle Q(j\omega) = m_3 \cdot \angle \left(\left(\frac{j\omega}{\omega_n} \right)^2 + 2j\zeta \left(\frac{\omega}{\omega_n} \right) + 1 \right)$$

$$t_g(\omega) = -m_3 \frac{d}{d\omega} \tan^{-1} \frac{2\zeta \left(\frac{\omega}{\omega_n} \right)}{1 - \frac{\omega^2}{\omega_n^2}}$$

$$t_g(\omega) = -m_3 \frac{2\zeta \omega_n (\omega^2 + \omega_n^2)}{\omega^4 + (4\zeta^2 - 2)\omega^2 \omega_n^2 + \omega_n^4}$$

$$t_g(\infty) = 0$$

$$t_g(\omega_n) = -m_3 \frac{2\zeta \omega_n (2\omega_n^2)}{(4\zeta^2 - 2)\omega_n^4 + 2\omega_n^4} = -m_3 \frac{4\zeta \omega_n^3}{4\zeta^2 \omega_n^4} = -m_3 \frac{1}{\zeta \omega_n}$$

$$t_g(0) = -m_3 \frac{2\zeta}{\omega_n} \quad \frac{t_g(\omega_n)}{t_g(0)} = \frac{1}{2\zeta^2}$$

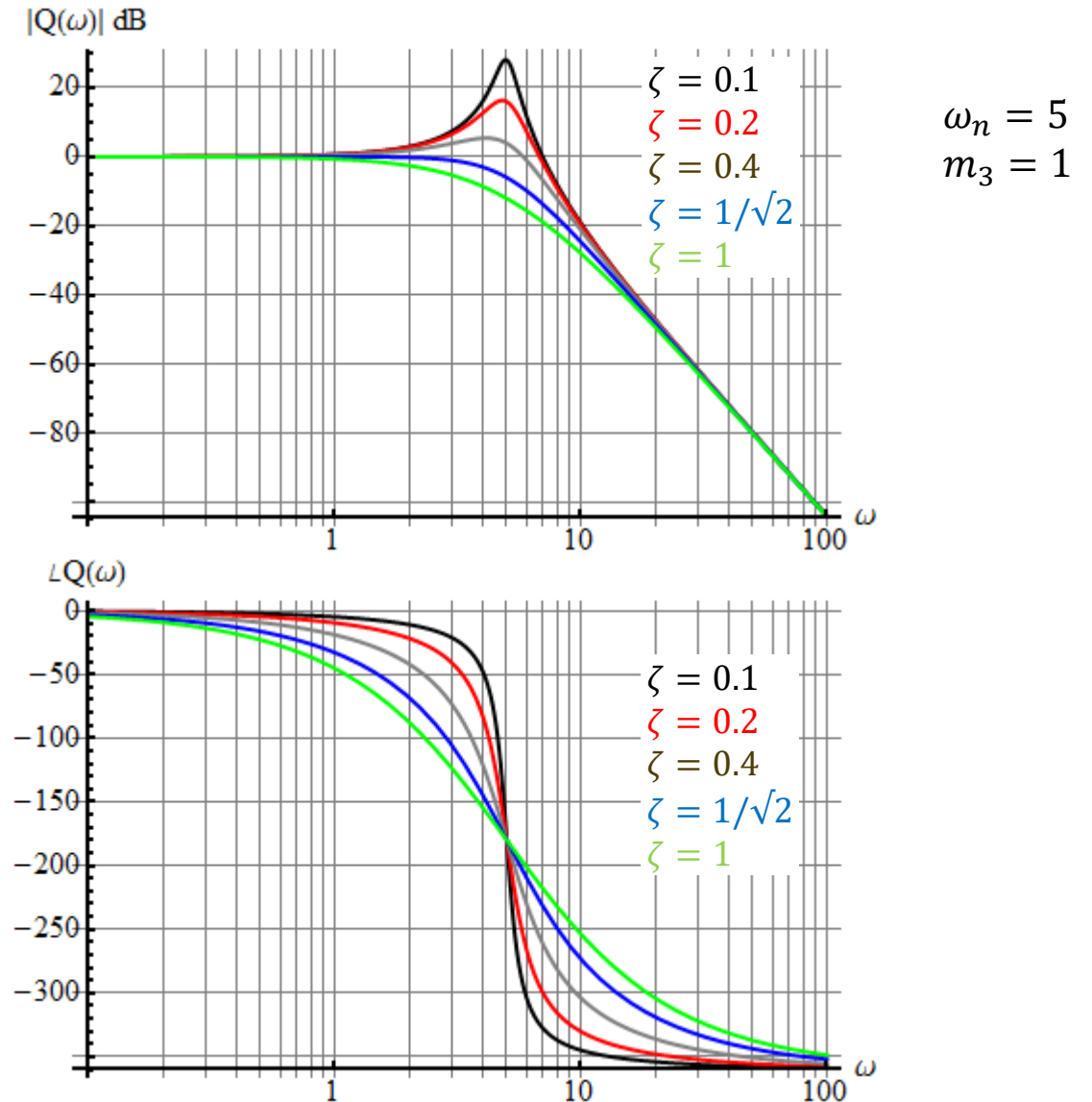
Single second order pole with different damping ratio

The effect of the damping ratio on the frequency characteristic.

If a system has very little damping it will tend to oscillate at a resonance frequency.

The phase transition band becomes very narrow for small damping ratios.

What is the filter order?



**Not in
Lathi's book!**

The low and high frequency boundaries for the phase transition are:

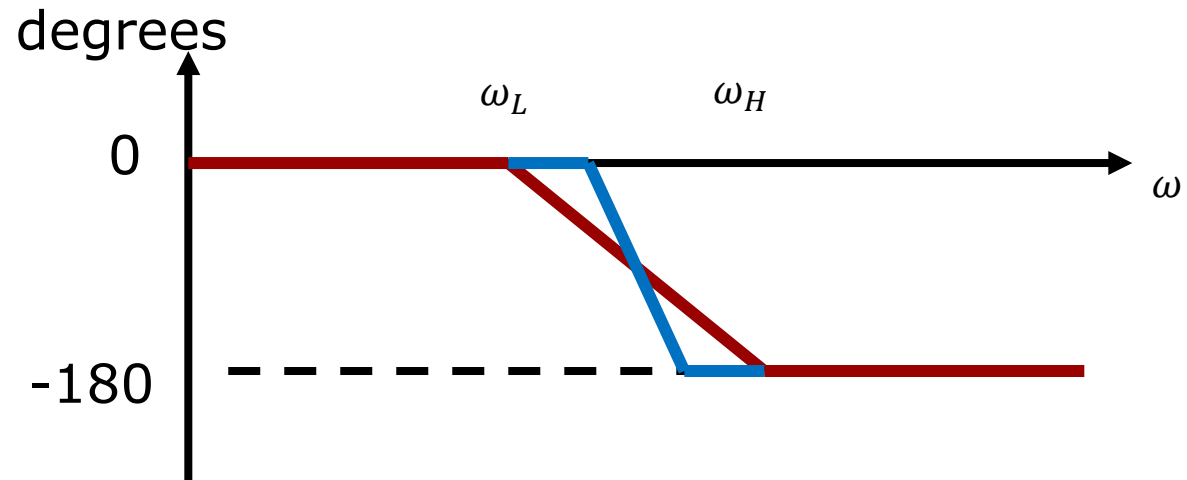
$$\omega_L = 10^{-a}\omega_n \wedge \omega_H = 10^a\omega_n$$

$$a = \begin{cases} 1.410\zeta - 0.150\zeta^2 & \zeta \leq 0.2 \\ 1.475\zeta - 0.475\zeta^2 & \zeta > 0.2 \end{cases}$$

Limits:

$$\begin{aligned} \zeta \rightarrow 1 &\Rightarrow \\ a \rightarrow 1 &\Rightarrow \\ \omega_L &\rightarrow \omega_n/10 \\ \omega_H &\rightarrow 10\omega_n \end{aligned}$$

$$\begin{aligned} \zeta \rightarrow 0 &\Rightarrow \\ \omega_L &\rightarrow \omega_n \\ \omega_H &\rightarrow \omega_n \end{aligned}$$



The blue system is more underdamped than the red.

Single Quadratic Factor

$$\left(\left(\frac{j\omega}{\omega_n} \right)^2 + 2j\zeta \frac{\omega}{\omega_n} + 1 \right)^{\pm 1}$$

Amplitude
slope:

$m_3 \cdot \frac{40\text{dB}}{\text{decade}}$
 $m_3 \cdot \frac{12\text{dB}}{\text{octave}}$

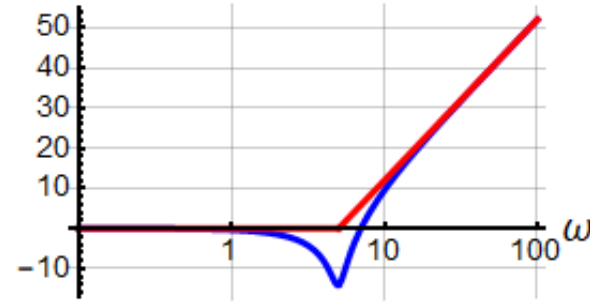
$$t_g(\omega_n) = -m_3 \frac{1}{\zeta \omega_n} = -2$$

$$t_g(0) = -m_3 \frac{2\zeta}{\omega_n} = -0.04$$

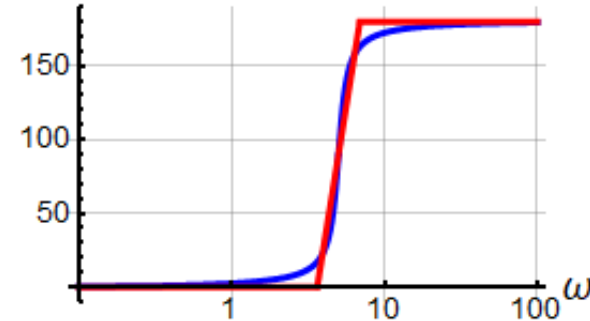
$$\frac{t_g(\omega_n)}{t_g(0)} = \frac{1}{2\zeta^2} = 50$$

$$m_3 = 1$$

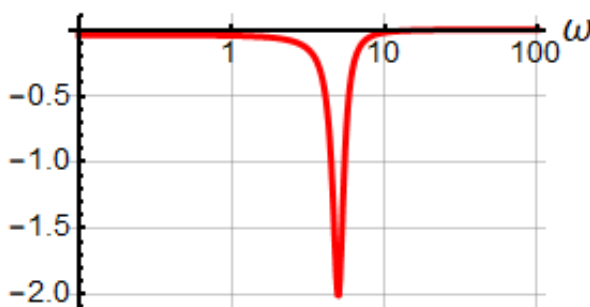
$|Q_1(\omega)|$ dB



$\angle Q_1(\omega)$

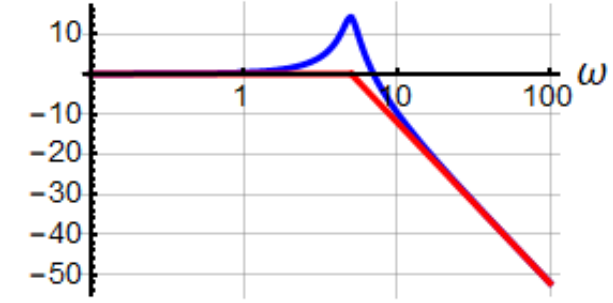


$t_{gQ1}(\omega)$

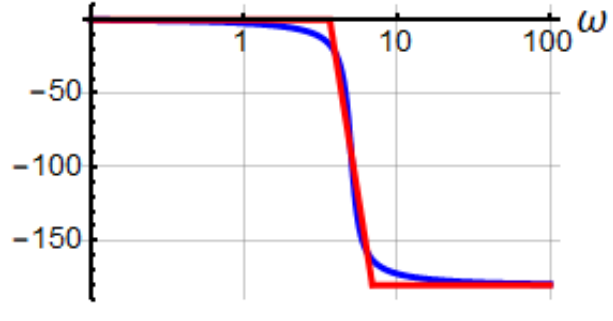


$$m_3 = -1$$

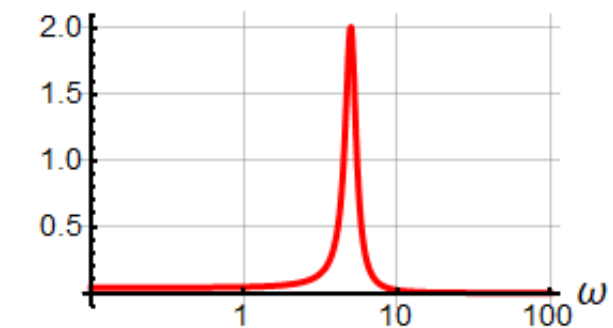
$|Q_2(\omega)|$ dB



$\angle Q_2(\omega)$



$t_{gQ2}(\omega)$



$$\omega_n = 5$$

$$\zeta = 0.1$$

$$m_3 = 1$$

Double Quadratic Factor

$$\left(\left(\frac{j\omega}{\omega_n} \right)^2 + 2j\zeta \frac{\omega}{\omega_n} + 1 \right)^{\pm 2}$$

Amplitude
slope:

$$m_3 \cdot \frac{40\text{dB}}{\text{decade}}$$

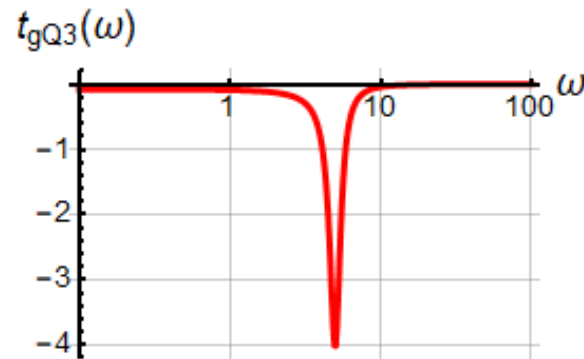
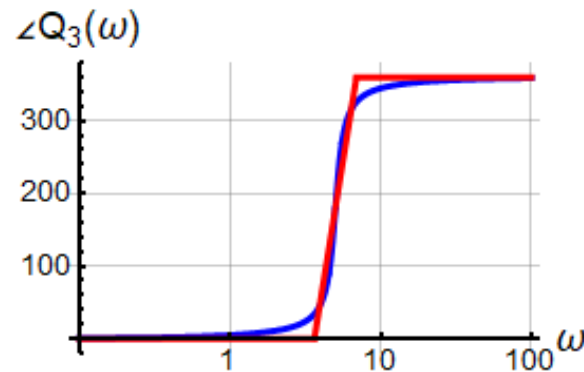
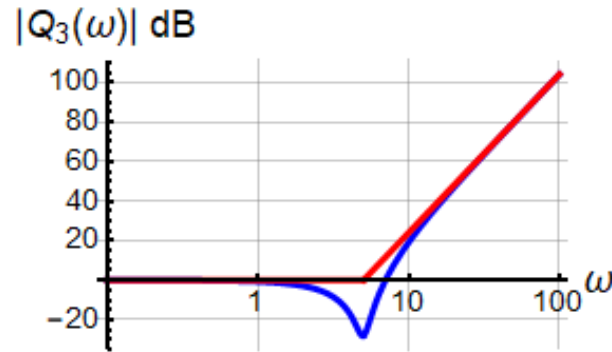
$$m_3 \cdot \frac{12\text{dB}}{\text{octave}}$$

$$t_g(\omega_n) = -m_3 \frac{1}{\zeta \omega_n} = -4$$

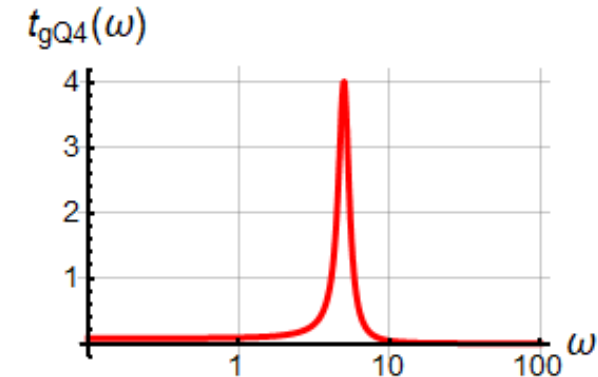
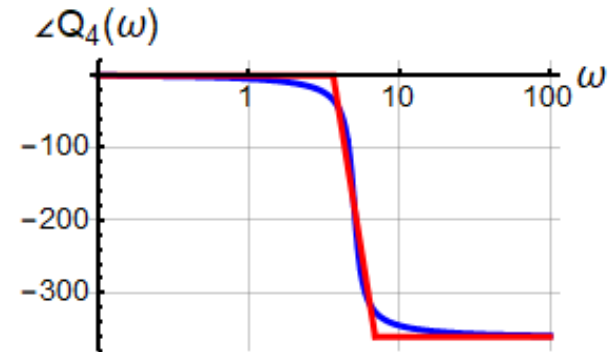
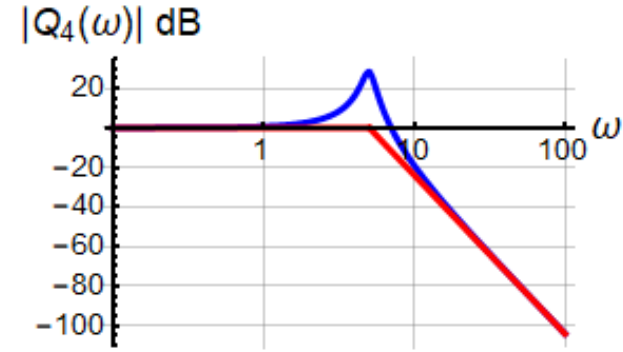
$$t_g(0) = -m_3 \frac{2\zeta}{\omega_n} = -0.08$$

$$\frac{t_g(\omega_n)}{t_g(0)} = \frac{1}{2\zeta^2} = 50$$

$$m_3 = 2$$



$$m_3 = -2$$



$$\omega_n = 5$$

$$\zeta = 0.1$$

$$m_3 = 2$$

Double Quadratic Factor

In the previous lecture we found the 3dB cut-off frequency for a single quadratic factor:

$$\omega_{3dB} = \omega_n \sqrt{1 - 2\zeta^2 + \sqrt{(4\zeta^4 - 4\zeta^2 + 2)}}$$

Here is used as an example:

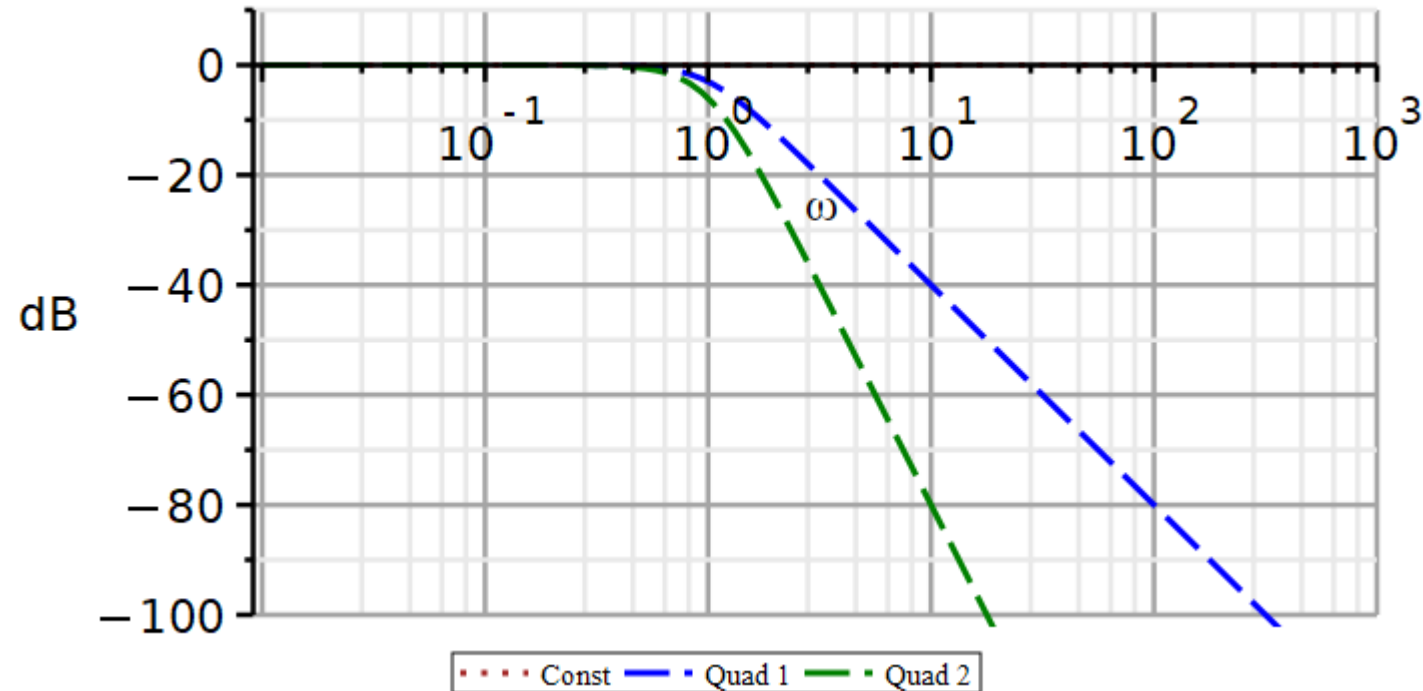
$$\zeta = \frac{1}{\sqrt{2}} \Rightarrow \omega_{3dB} = \omega_n = 1$$

The blue curve is a *single* quadratic factor ($m_3 = -1$). Hence, the amplitude at ω_n is **-3dB**.

The green curve is a *double* quadratic factor ($m_3 = -2$). Hence, the amplitude at ω_n is **-6dB**.

$$|Q(j\omega)|_{dB} = m_3 \cdot 20 \log \left| \left(\frac{j\omega}{1} \right)^2 + 2 \frac{1}{\sqrt{2}} \left(\frac{j\omega}{1} \right) + 1 \right|$$

$$|Q(j\omega_{3dB})|_{dB} = m_3 \cdot (3dB)$$



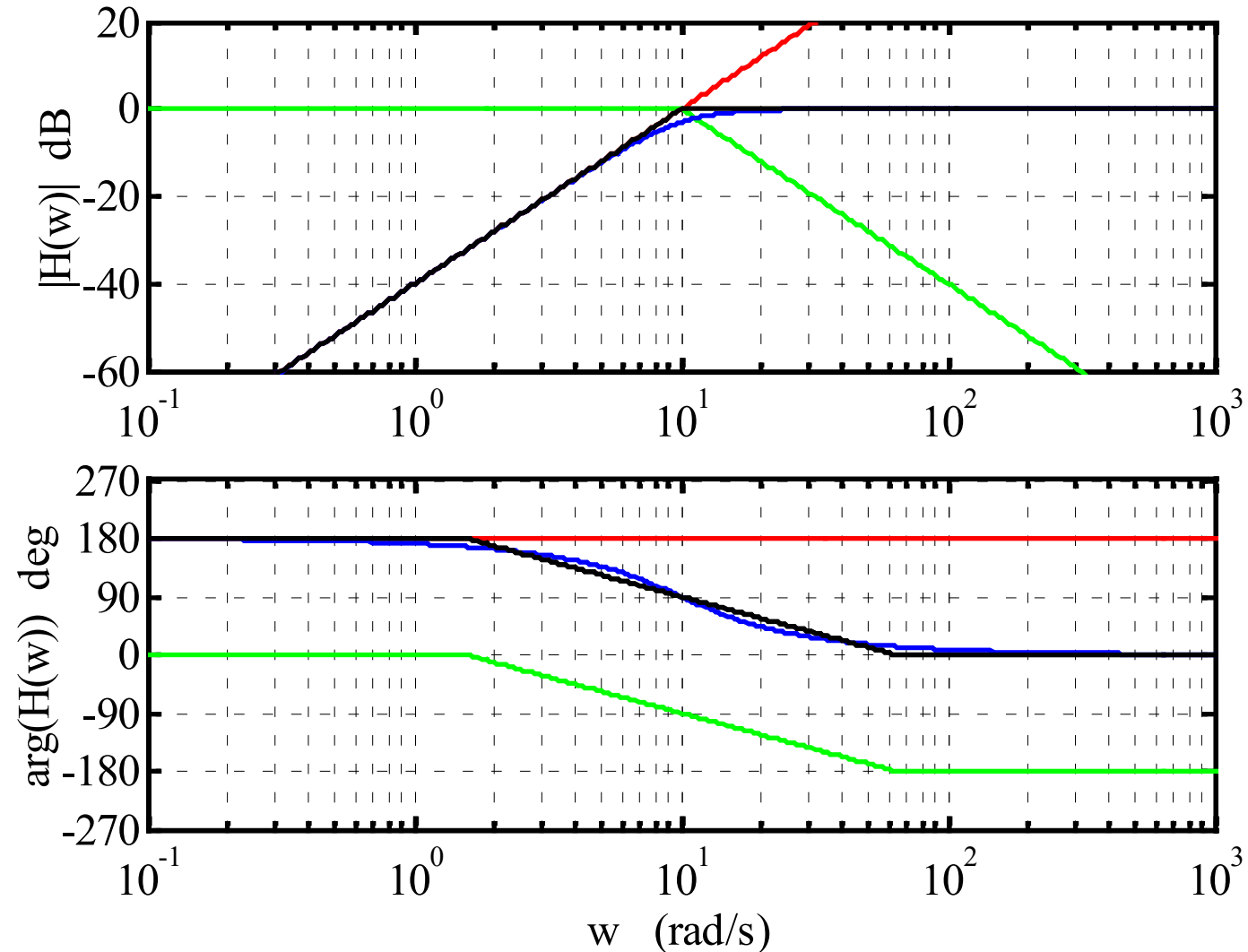
Bode Plot example

Analyze the Bode plot and identify the transfer function.

$$\Omega(\omega) = \left(\frac{j\omega}{10}\right)^2$$

$$Q(\omega) = \left(\left(\frac{j\omega}{10}\right)^2 + 2\zeta\frac{j\omega}{10} + 1\right)^{-1}$$

Why not: $\left(\frac{j\omega}{10} + 1\right)^{-2}$



Solution: Bode Plot example

Red curve:

$$\Omega(\omega) = \left(\frac{j\omega}{10}\right)^2$$

Green curve:

$$Q(\omega) = \left(\left(\frac{j\omega}{10}\right)^2 + 2\zeta\frac{j\omega}{10} + 1\right)^{-1}$$

At what frequency does the phase transition stop:

$$\omega_H = 10 \cdot 10^a = 60 \Rightarrow a = \log_{10}(6) = 0.778$$

Solve for ζ

$$a = 1.475\zeta - 0.475\zeta^2 = 0.778 \Rightarrow \zeta = 0.67 \vee 2.43$$

The curves were constructed using:

$$\zeta = \frac{1}{\sqrt{2}} \approx 0.707$$

$$\begin{aligned} H(\omega) &= \frac{(j\omega/10)^2}{(j\omega/10)^2 + 2\zeta(j\omega/10) + 1} \\ &= \frac{(j\omega)^2}{(j\omega)^2 + 2 \cdot 0.67 \cdot 10j\omega + 100} \end{aligned}$$

$$H(s) = \frac{s^2}{(s^2 + 2 \cdot 0.67 \cdot 10 \cdot s + 100)}$$

For underdamped 2nd order systems with a local peak in the amplitude spectrum, we can obtain estimates of $\zeta \wedge \omega_n$ from the amplitude peak and peak frequency:

$$M_{pk} = m_3 \cdot 20 \cdot \log \left[2\zeta \sqrt{1 - \zeta^2} \right] dB$$

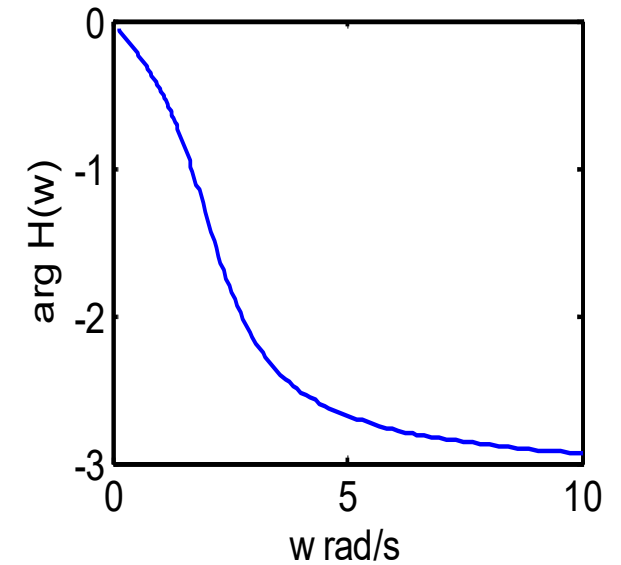
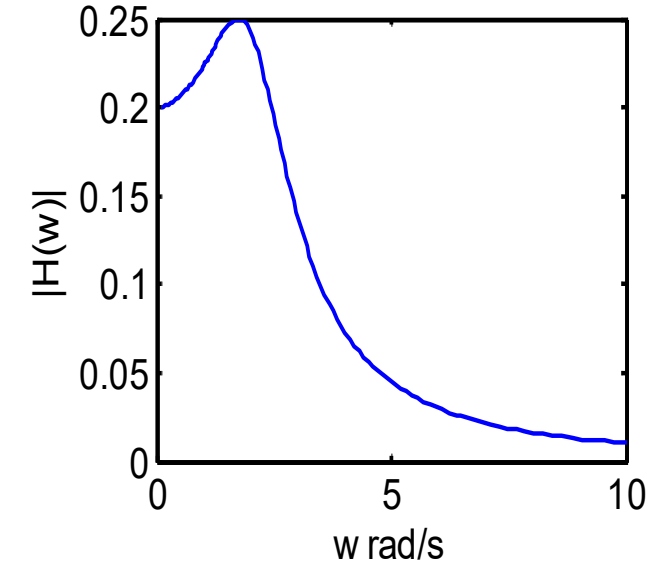
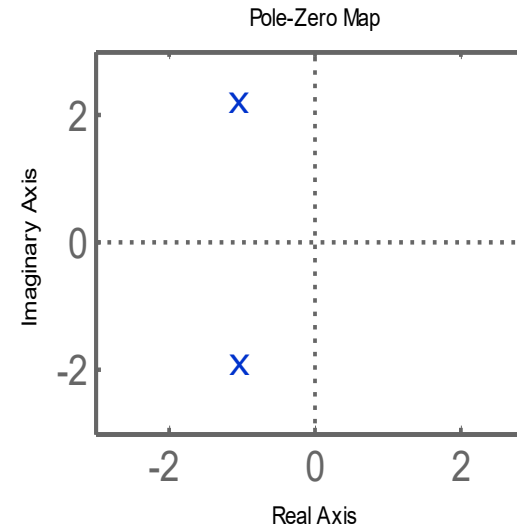
$$\omega_{pk} = \omega_n \sqrt{1 - 2\zeta^2}$$

Knowing $\zeta \wedge \omega_n$ we have defined the differential equation, and we can estimate settling time, percent overshoot, delay time, time to peak, and rise time of the step response.

Filter design by placement of poles and zeros

Video

How can we explain the frequency characteristic from the locations of the poles and zeros?



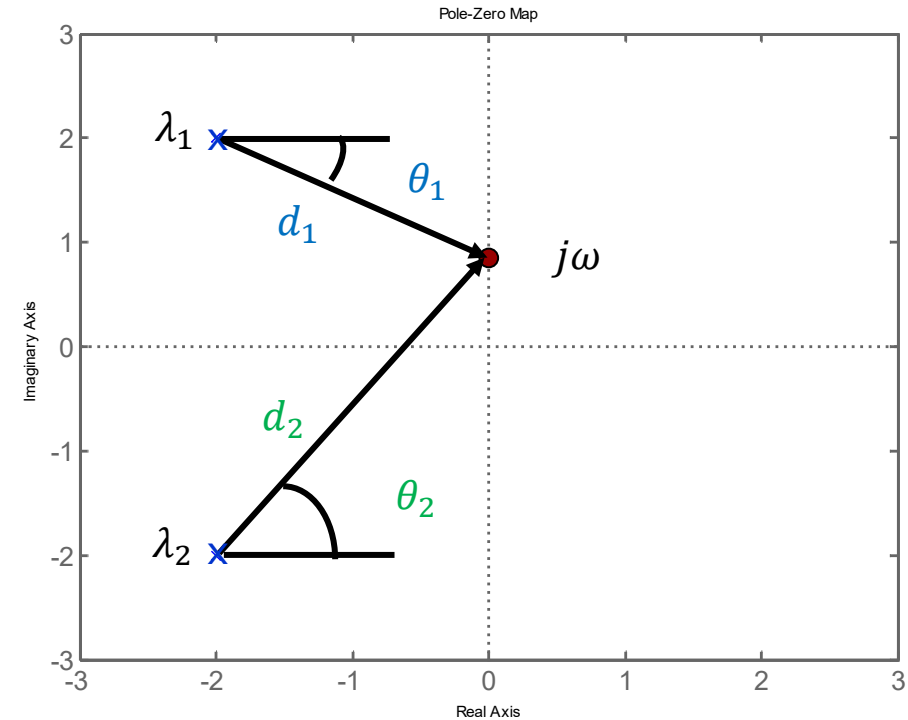
Graphical Interpretation

Here we define the transfer function in terms of its poles $(\lambda_1, \lambda_2 \dots)$.

$$H(s) = \frac{1}{(s - \lambda_1)(s - \lambda_2)}$$

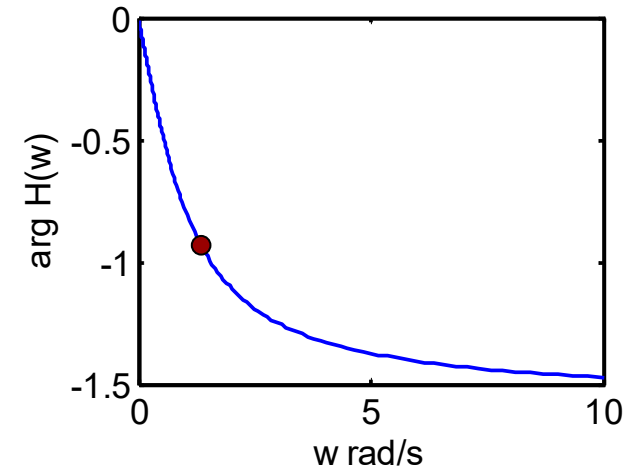
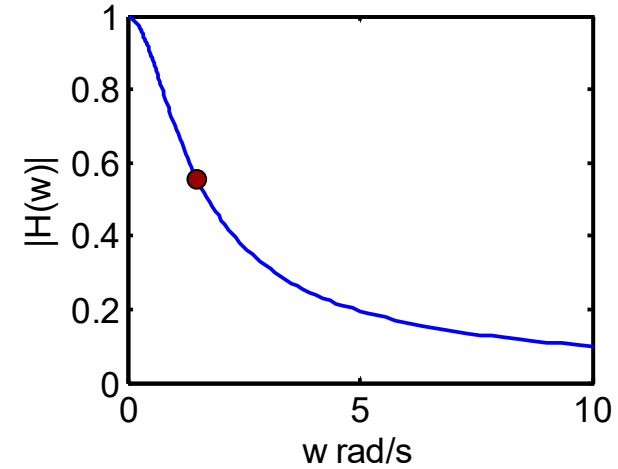
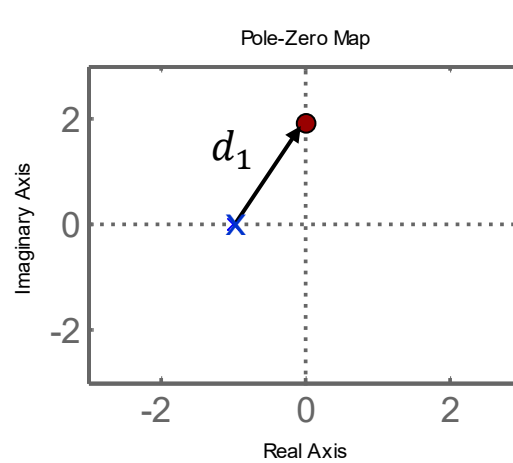
$$\begin{aligned} H(s) \Big|_{s=j\omega} &= \frac{1}{(j\omega - \lambda_1)(j\omega - \lambda_2)} \\ &= \underbrace{\frac{1}{d_1 d_2}}_{\text{polar form}} e^{-j(\theta_1 + \theta_2)} \end{aligned}$$

d_1 and d_2 are the distance from the poles to a location $(j\omega)$ on the imaginary axis.
The angles are as drawn in the diagram.



For positive frequencies
the angle is positive
but the factor is in the
denominator, thus the
phase is negative.

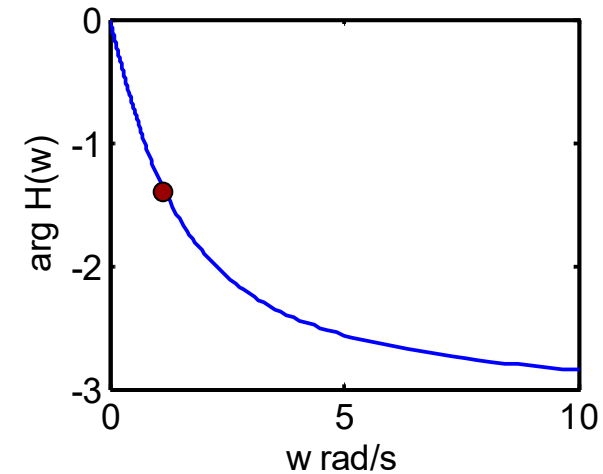
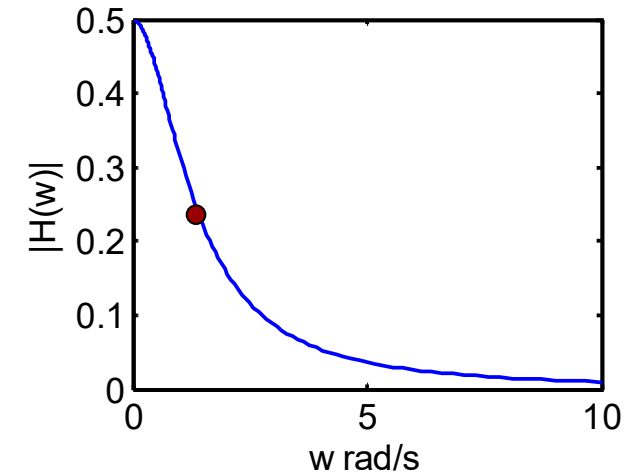
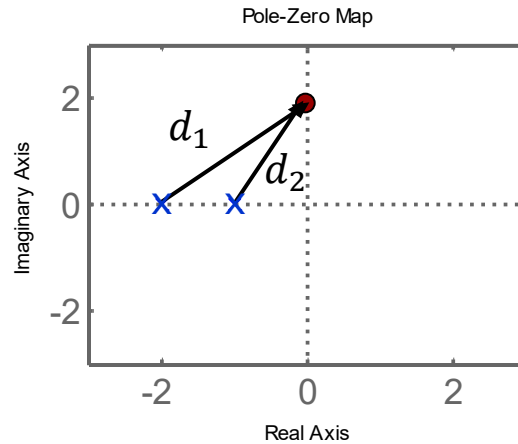
$$\begin{aligned}
 H(s) \Big|_{s=j\omega} &= \frac{1}{(j\omega - \lambda_1)} \\
 &= \underbrace{\frac{1}{d_1}}_{\text{polar form}} e^{-j\theta_1}
 \end{aligned}$$



Two real poles

We now have two angles which approach 90.
Thus, the phase settles at -180 degrees.

$$\begin{aligned}
 H(s) \Big|_{s=j\omega} &= \frac{1}{(j\omega - \lambda_1)(j\omega - \lambda_2)} \\
 &= \underbrace{\frac{1}{d_1 d_2}}_{\text{polar form}} e^{-j(\theta_1 + \theta_2)}
 \end{aligned}$$



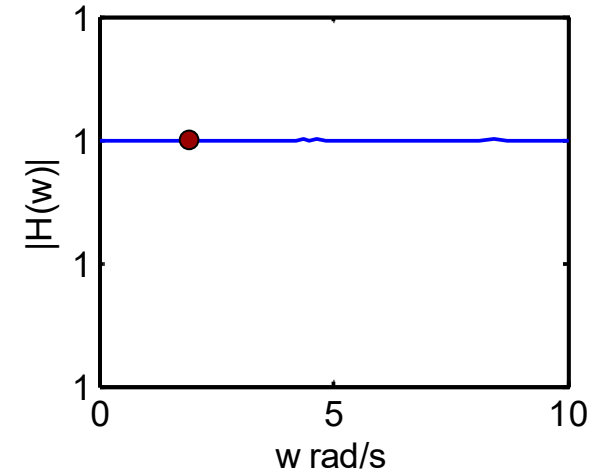
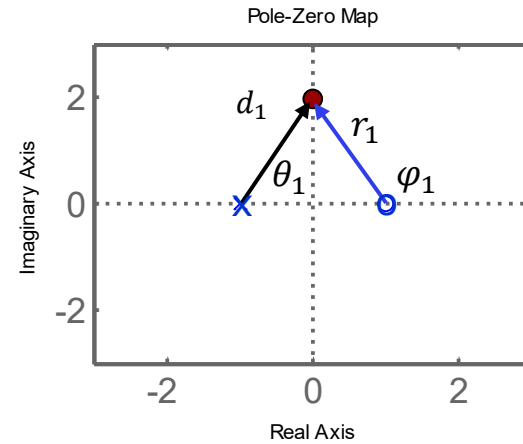
One pole and one positive zero

Amplitude
response:

$$\frac{\prod_m r_m}{\prod_n d_n}$$

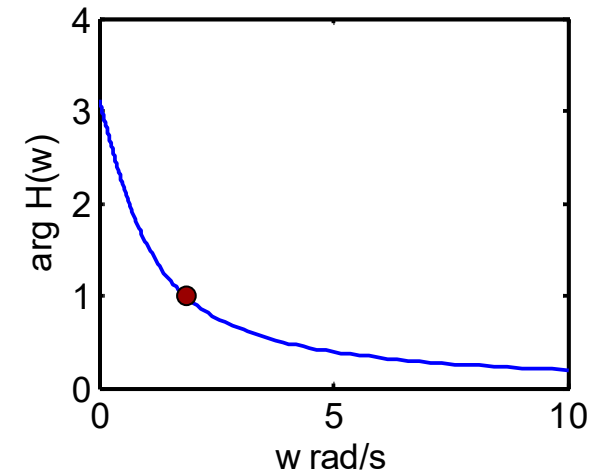
Phase
response:

$$\sum_m \varphi_m - \sum_n \theta_n$$



Why is the
amplitude
spectrum
constant?

$$H(s) \Big|_{s=j\omega} = \frac{r_1}{d_1} \cdot e^{j(\varphi_1 - \theta_1)}$$



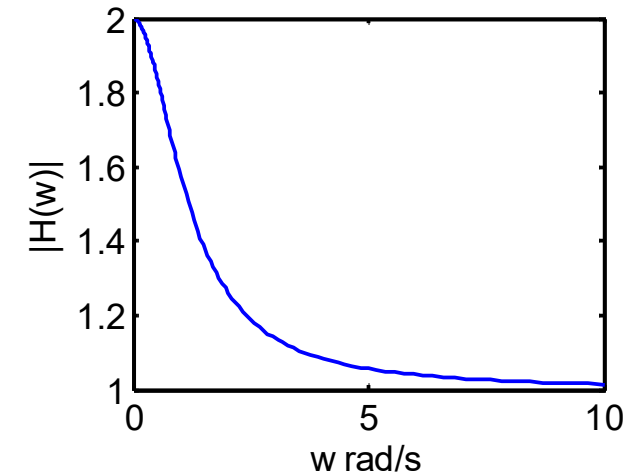
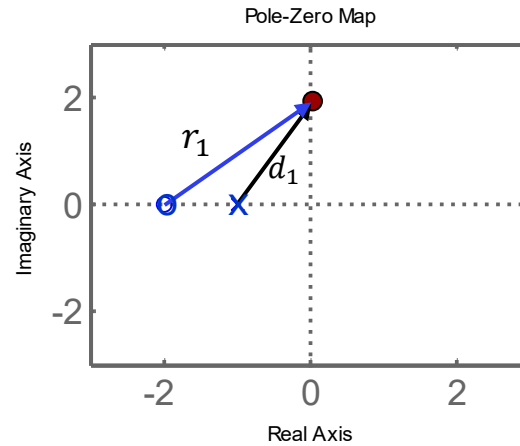
One pole and one negative zero

Amplitude
response:

$$\frac{\prod_m r_m}{\prod_n d_n}$$

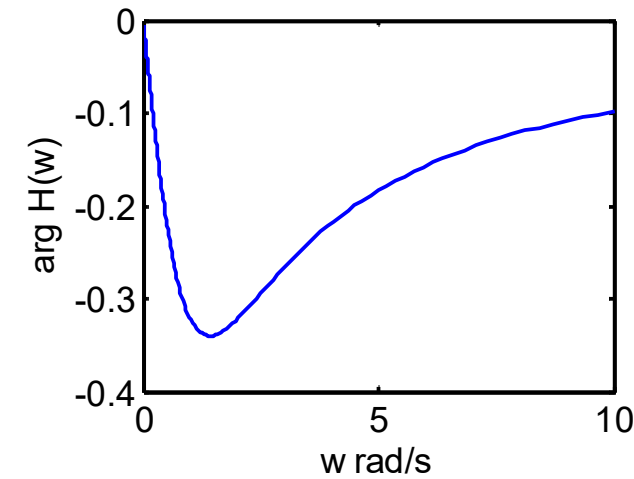
Phase
response:

$$\sum_m \varphi_m - \sum_n \theta_n$$



← Why 2?

← Why 1?



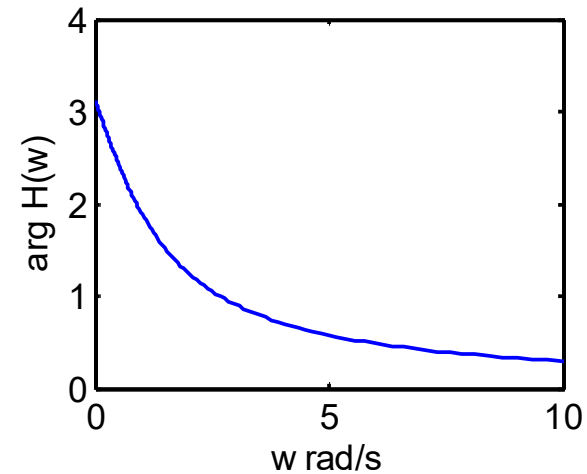
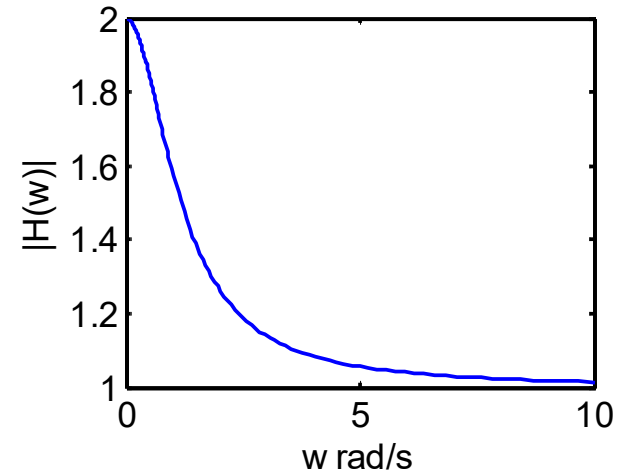
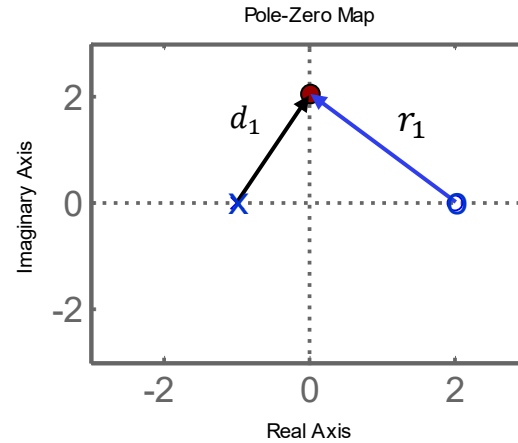
Why does the phase angle vary much less in this case than in the previous case?

Amplitude
response:

$$\frac{\prod_m r_m}{\prod_n d_n}$$

Phase
response:

$$\sum_m \varphi_m - \sum_n \theta_n$$



What effect does the sign
of the zero have on the
amplitude response?
Compare with previous
slide.

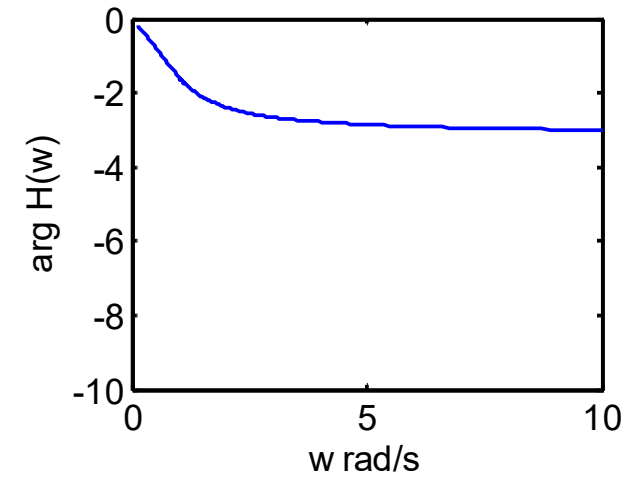
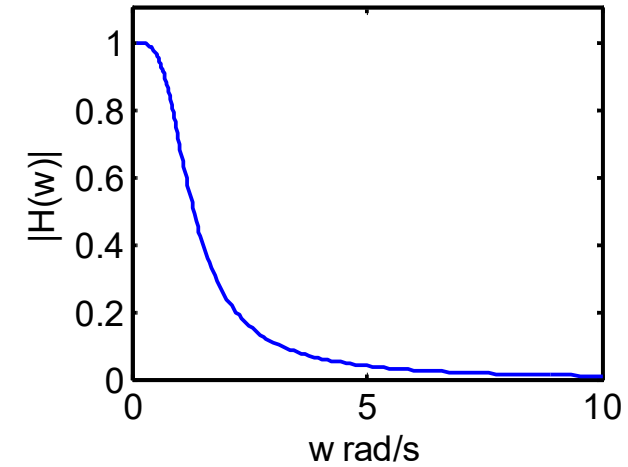
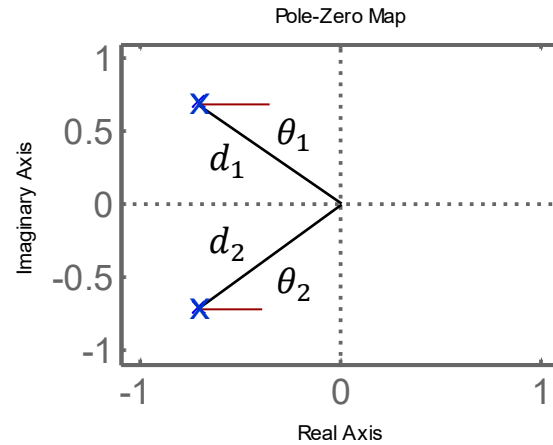
2nd order lowpass filter

Amplitude
response:

$$\frac{1}{\prod_n d_n}$$

Phase
response:

$$0 - \sum_n \theta_n$$



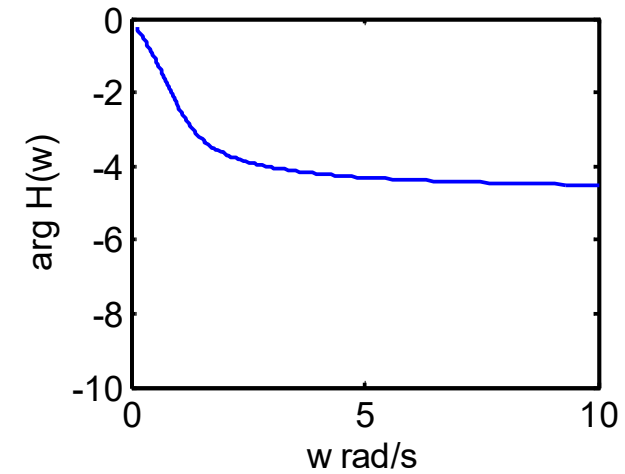
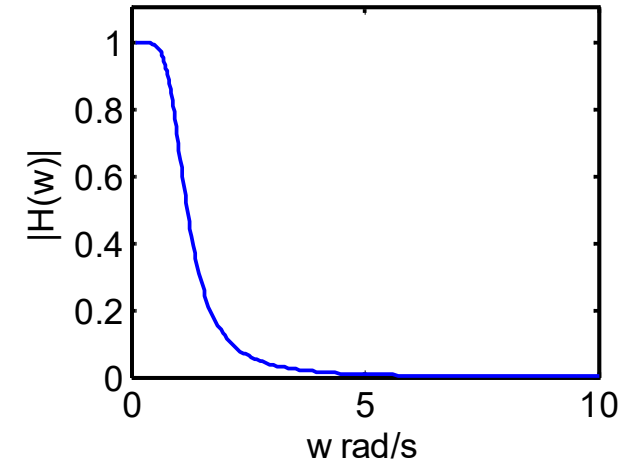
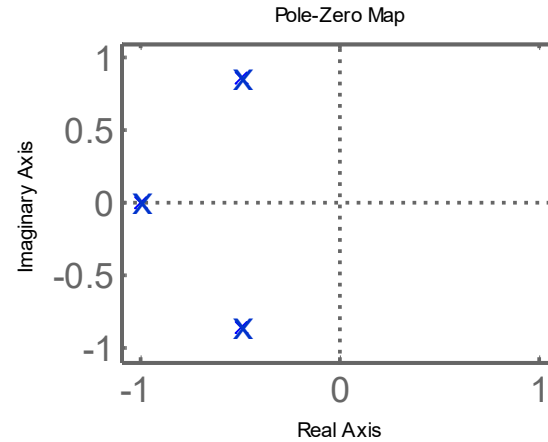
3rd order lowpass filter

Amplitude
response:

$$\frac{1}{\prod_n d_n}$$

Phase
response:

$$0 - \sum_n \theta_n$$



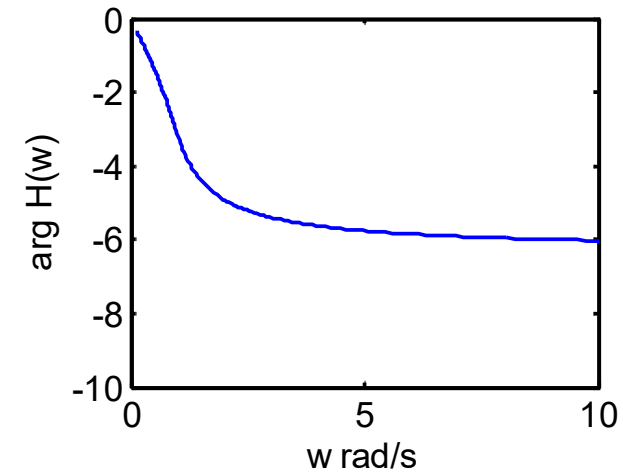
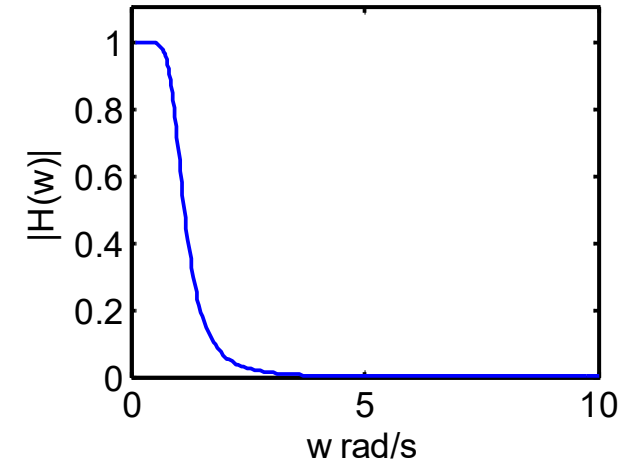
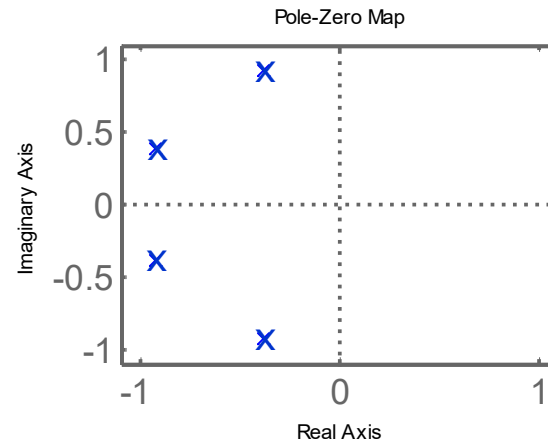
4th order lowpass filter

Amplitude
response:

$$\frac{1}{\prod_n d_n}$$

Phase
response:

$$0 - \sum_n \theta_n$$



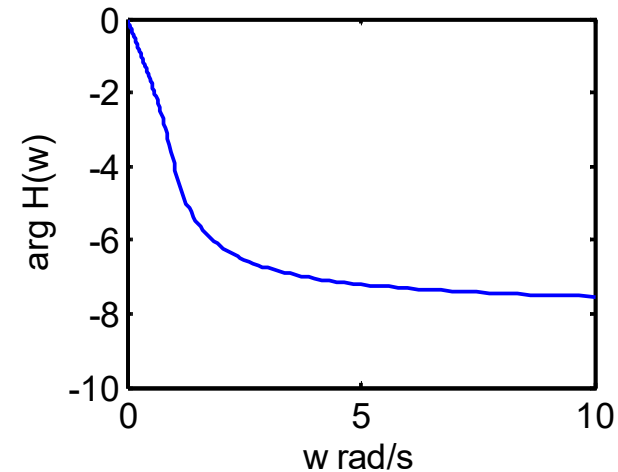
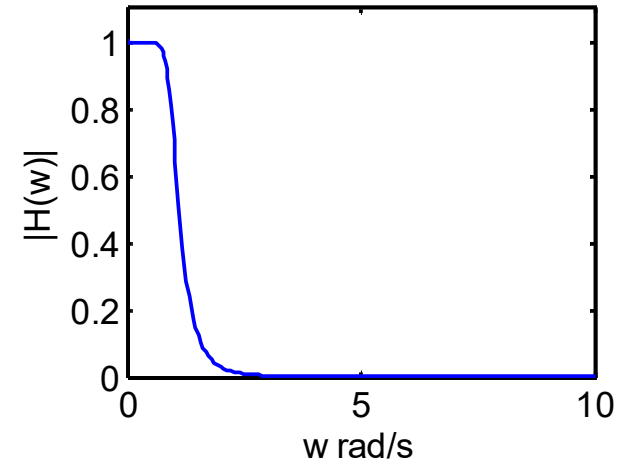
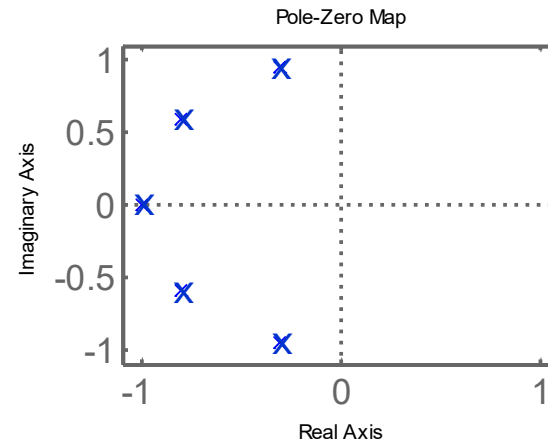
5th order lowpass filter

Amplitude
response:

$$\frac{1}{\prod_n d_n}$$

Phase
response:

$$0 - \sum_n \theta_n$$



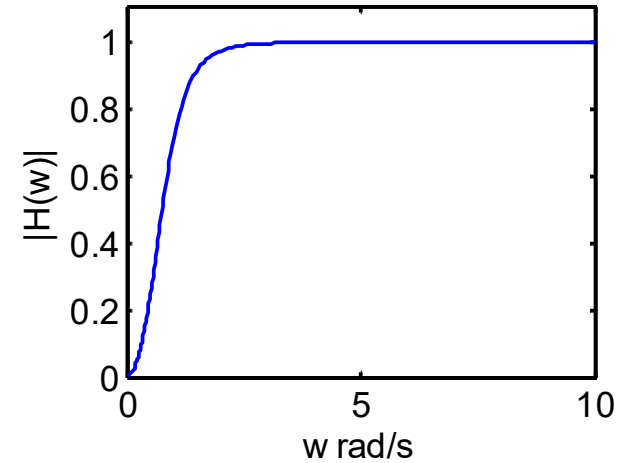
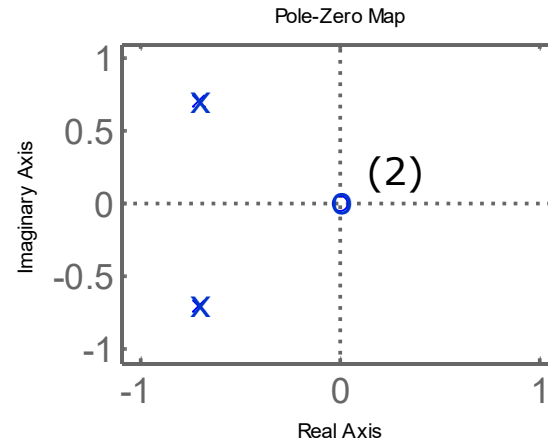
Notice that filters of odd orders have poles on the negative real axis.

All these filters have poles on the unit circle and are of the **Butterworth type**.

2nd order highpass filter

Amplitude
response:

$$\frac{\prod_m r_m}{\prod_n d_n}$$

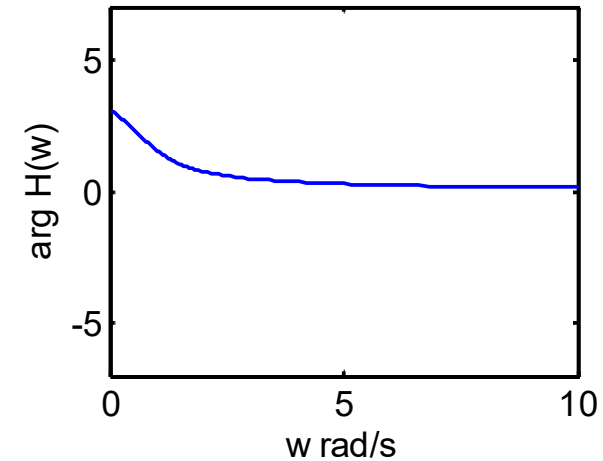


Phase
response:

$$\sum_m \varphi_m - \sum_n \theta_n$$

Why does the amplitude
go to zero in the origin?

Why does the phase
response jump from -180
to 180 degrees?



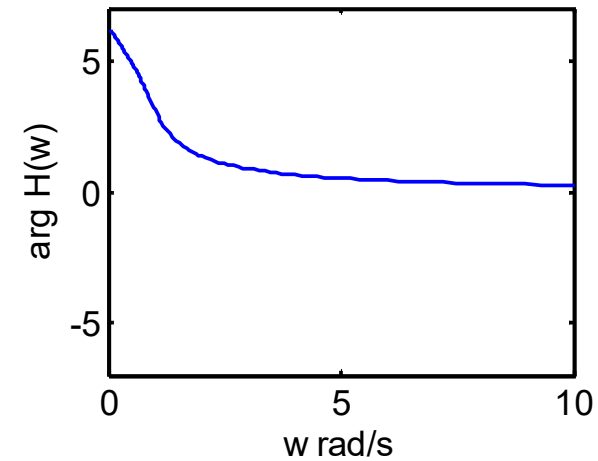
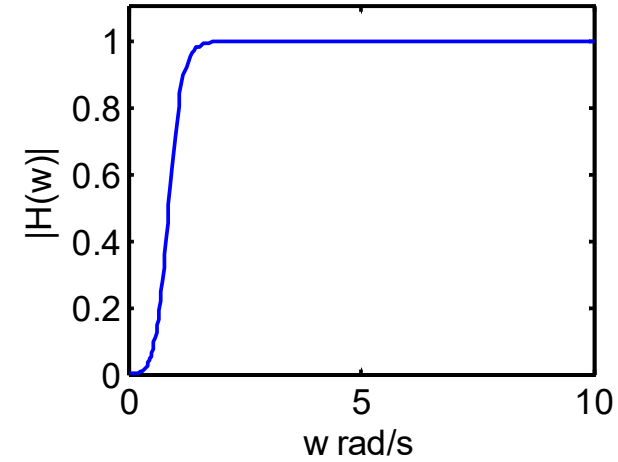
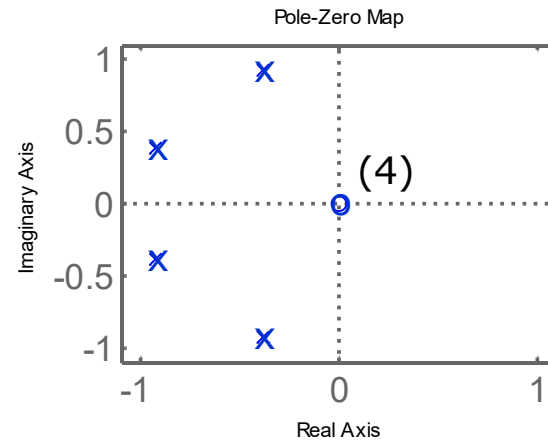
4th order highpass filter

Amplitude
response:

$$\frac{\prod_m r_m}{\prod_n d_n}$$

Phase
response:

$$\sum_m \varphi_m - \sum_n \theta_n$$



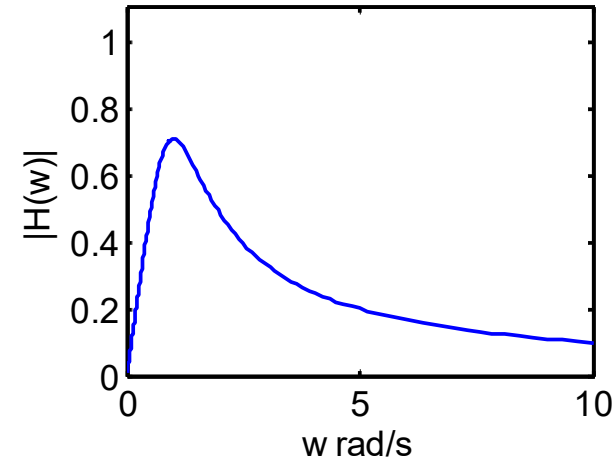
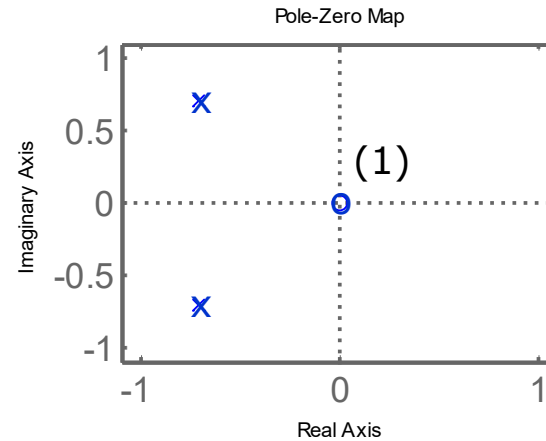
What is the phase jump
in 0 rad/s?

$$4 \cdot 180^\circ = 720^\circ$$

2nd order bandpass filter

Amplitude
response:

$$\frac{\prod_m r_m}{\prod_n d_n}$$

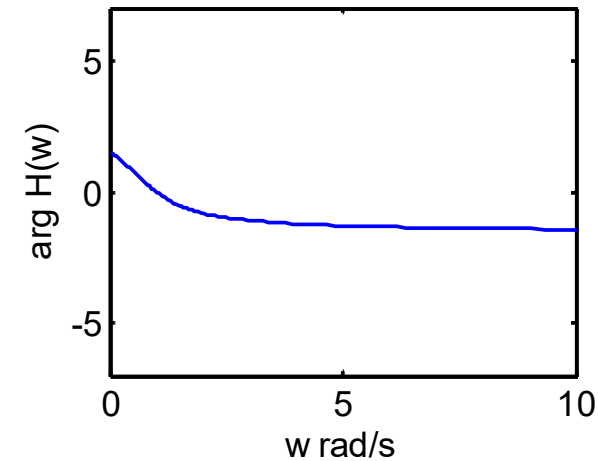


As $\omega \rightarrow \infty$ how fast does the
magnitude decrease, 1st or 2nd
order?

Phase
response:

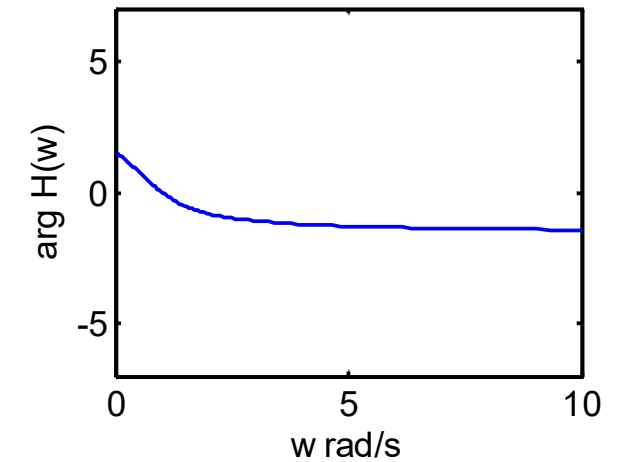
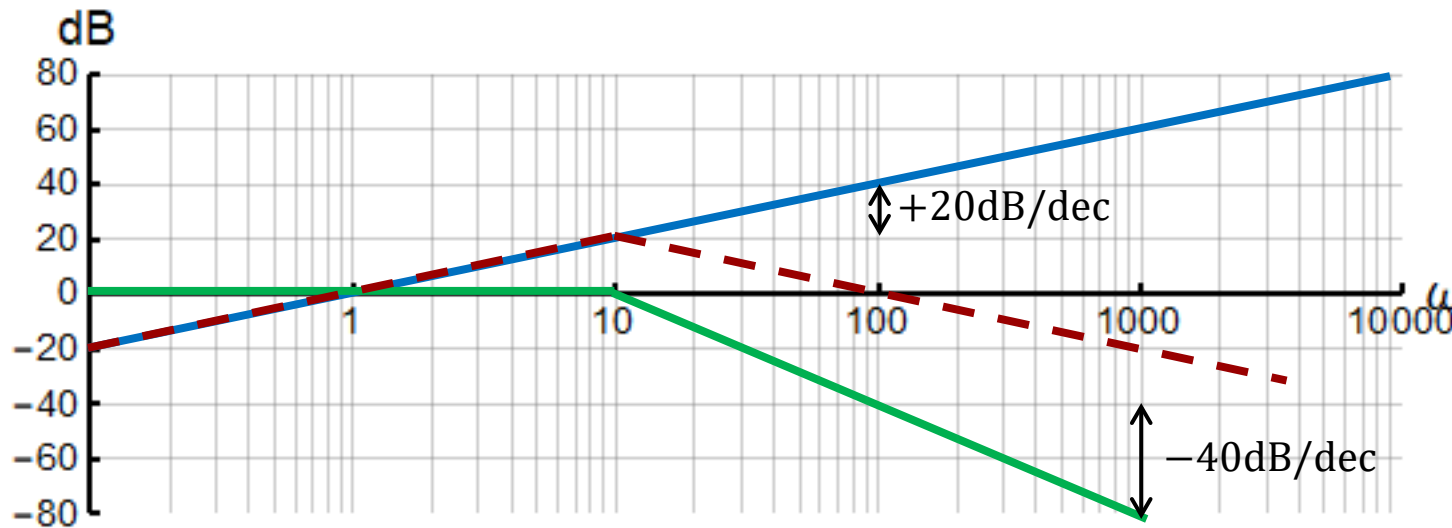
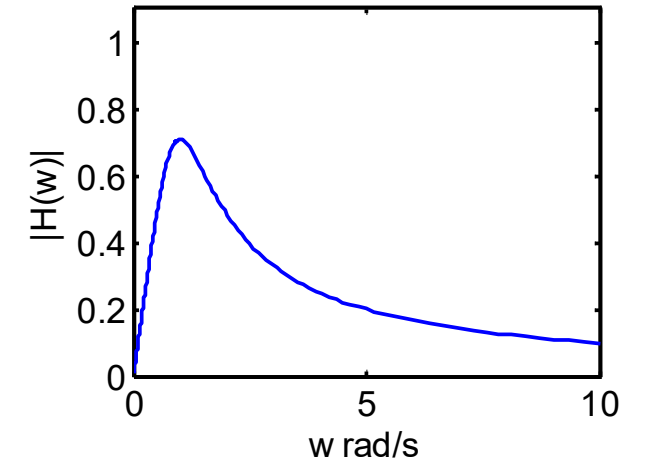
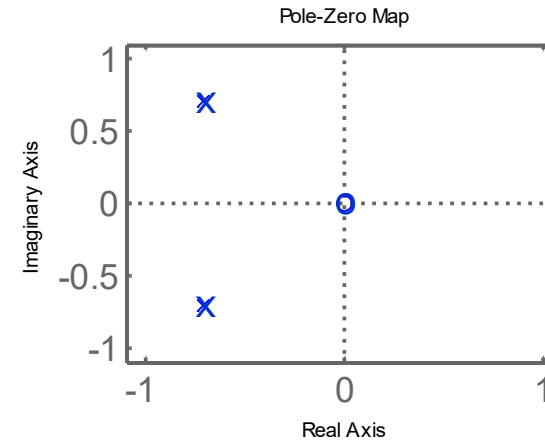
$$\sum_m \varphi_m - \sum_n \theta_n$$

What is the phase jump
in 0 rad/s?



$$1 \cdot 180^\circ = 180^\circ$$

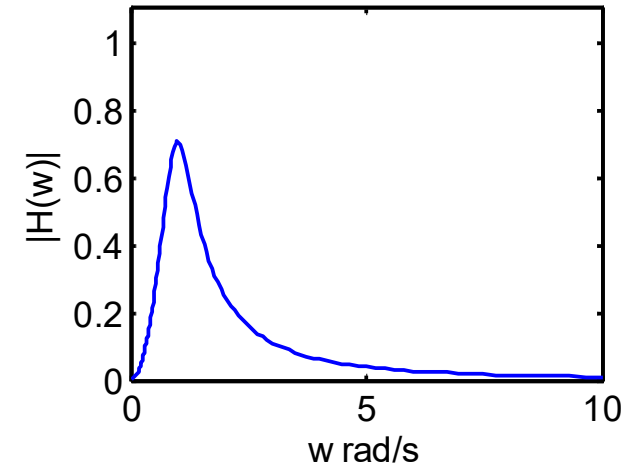
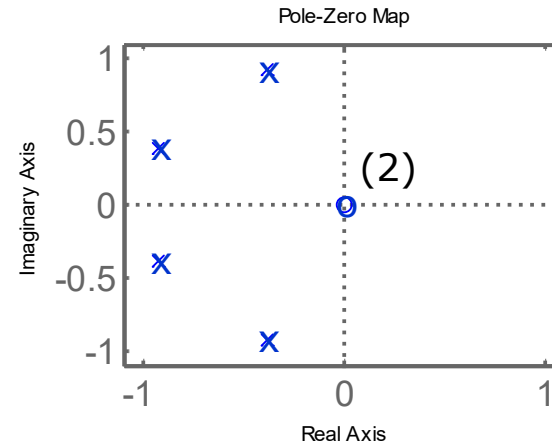
Does this amplitude decay agree with Bode plot?



4th order bandpass filter

Amplitude
response:

$$\frac{\prod_m r_m}{\prod_n d_n}$$



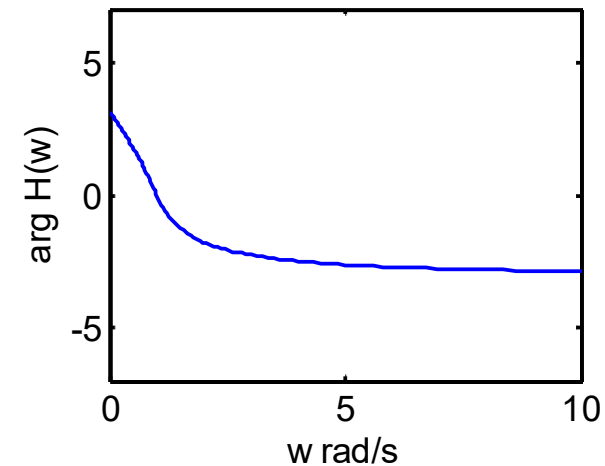
As $\omega \rightarrow \infty$ how fast does the
magnitude decrease, 1st or 2nd
order?

Phase
response:

$$\sum_m \varphi_m - \sum_n \theta_n$$

What is the phase jump
in 0 rad/s?

$$2 \cdot 180^\circ = 360^\circ$$



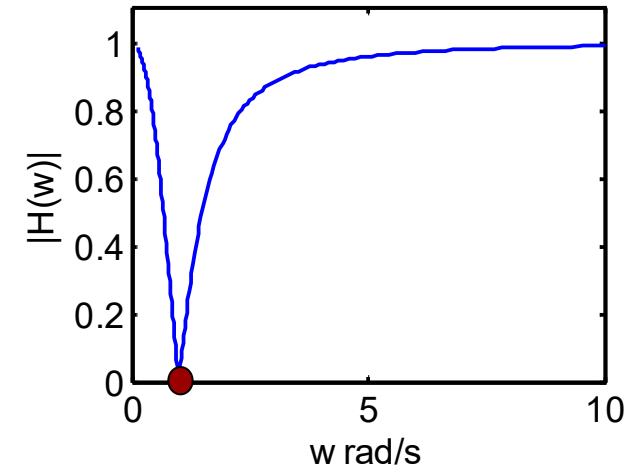
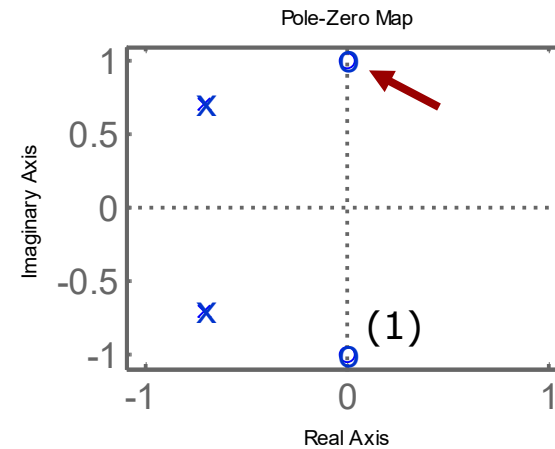
2nd order bandstop filter

Amplitude
response:

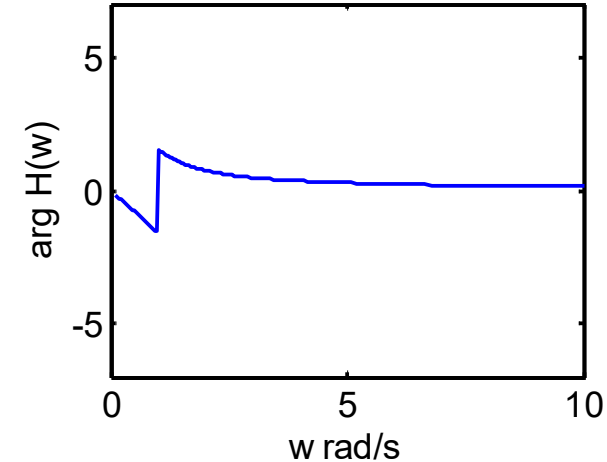
$$\frac{\prod_m r_m}{\prod_n d_n}$$

Phase
response:

$$\sum_m \varphi_m - \sum_n \theta_n$$



What creates the dip in
amplitude?



What is the phase jump
in 1 rad/s?

$$1 \cdot 180^\circ = 180^\circ$$

Twin-T notch filter

$$H(s) = \frac{(RCs + 1)((RC)^2s^2 + 1)}{(RCs + 1)((RC)^2s^2 + 4(1 - \alpha)(RC)s + 1)}$$

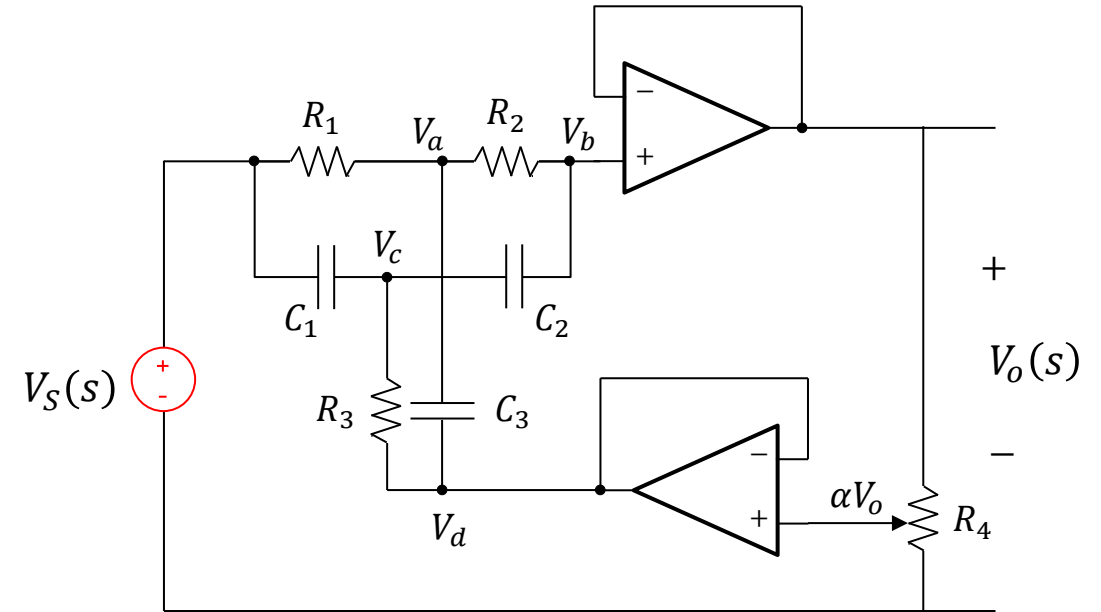
We like to study the poles of the quadratic factor.

$$((RC)^2s^2 + 4(1 - \alpha)(RC)s + 1) = 0$$

We get:
$$s_{1,2} = \frac{-2 + 2\alpha \pm \sqrt{4\alpha^2 - 8\alpha + 3}}{RC}$$

$$R = 68\text{k}\Omega \quad C = 47\text{nF}$$

Inserting numbers in the expression for the poles:



$$\alpha = 0.2: \quad s_{1,2} = -109.8, \quad -891.4$$

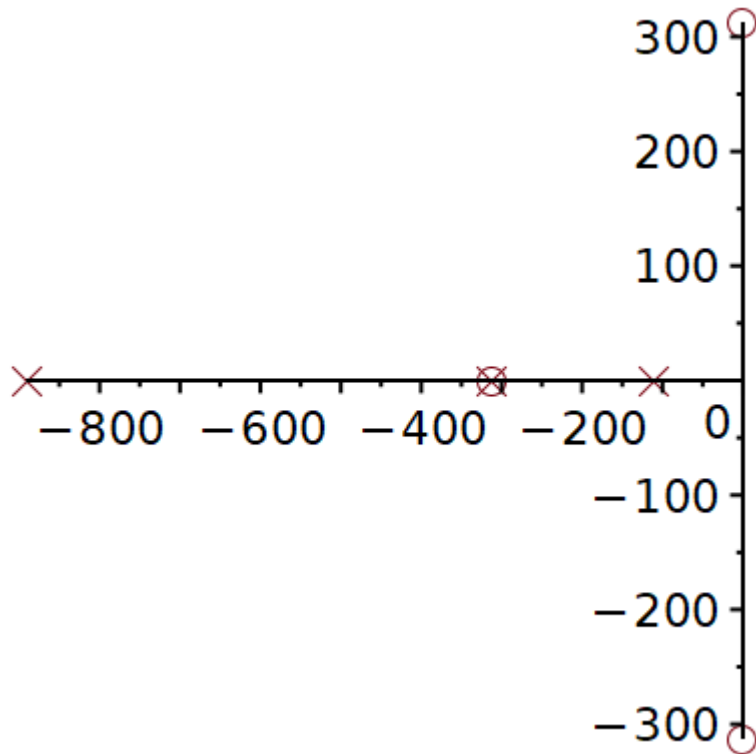
$$\alpha = 0.5: \quad s_{1,2} = -312.9, \quad -312.9$$

$$\alpha = 0.8: \quad s_{1,2} = -125.2 \pm j286.8$$

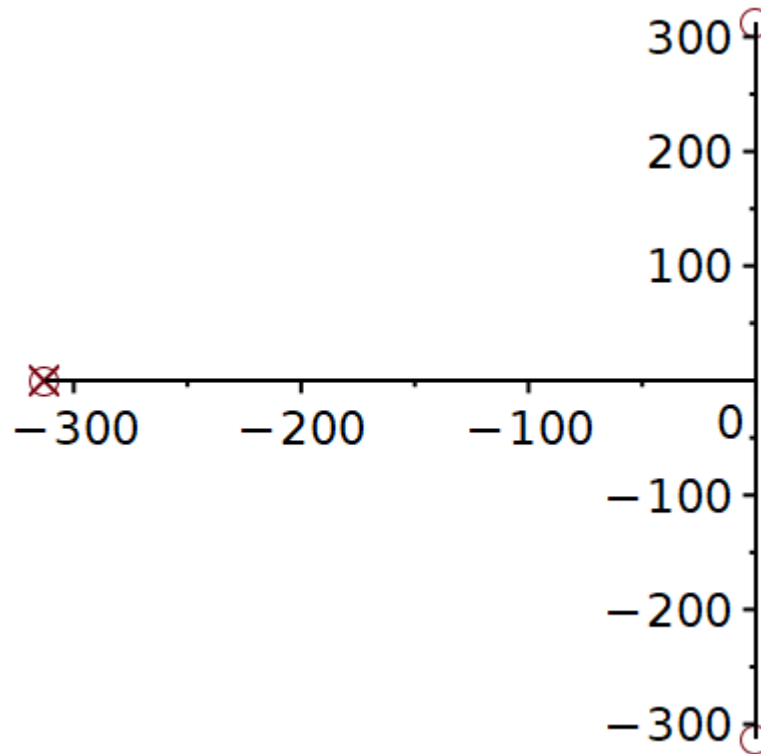
Twin-T notch filter

$$H(s) = \frac{(RCs + 1)((RC)^2s^2 + 1)}{(RCs + 1)((RC)^2s^2 + 4(1 - \alpha)(RC)s + 1)}$$

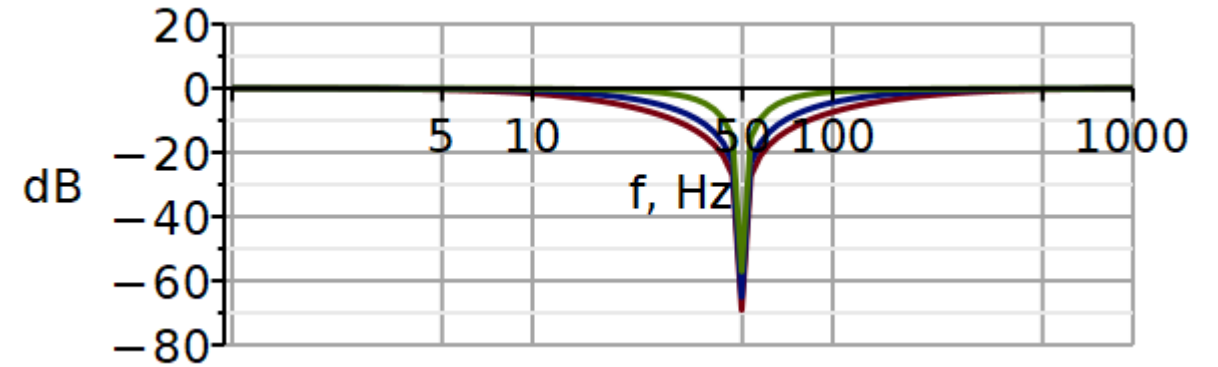
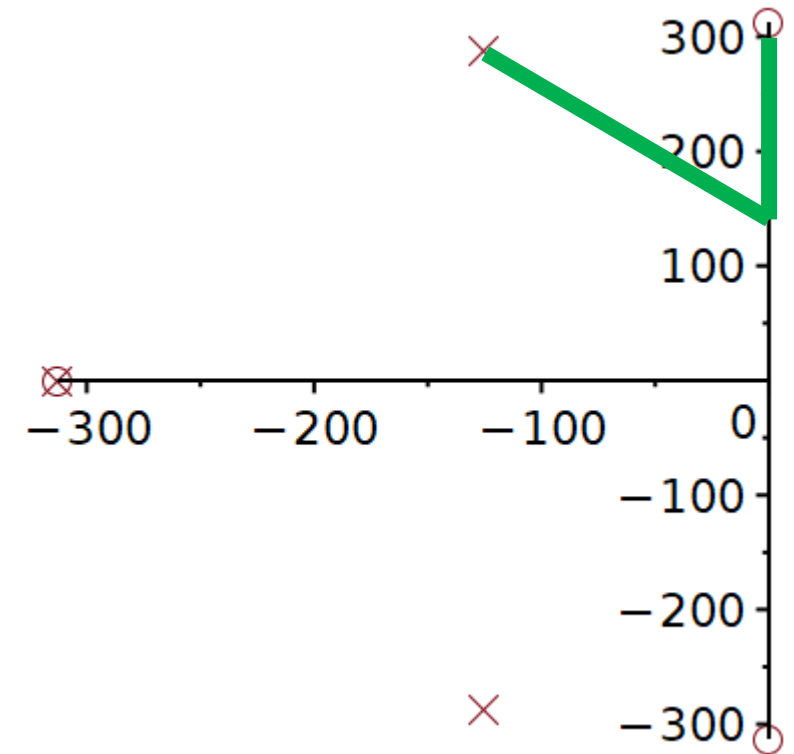
$$\alpha = 0.2, \zeta = 1.6$$



$$\alpha = 0.5, \zeta = 1$$



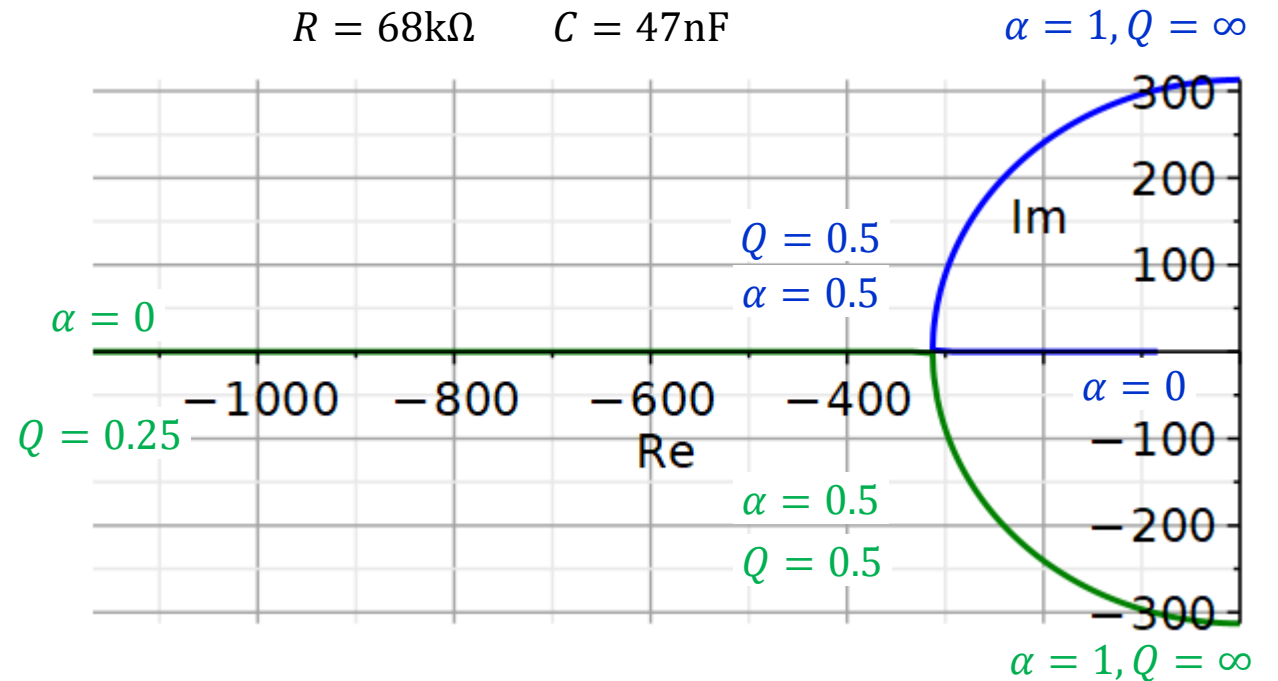
$$\alpha = 0.8, \zeta = 0.4$$



It is interesting to plot the **pole trajectories** when α is adjusted.

$$f_1(\alpha, R, C) = \frac{-2 + 2\alpha + \sqrt{4\alpha^2 - 8\alpha + 3}}{RC}$$

$$f_2(\alpha, R, C) = \frac{-2 + 2\alpha - \sqrt{4\alpha^2 - 8\alpha + 3}}{RC}$$



The adjustment of α only changes the damping factor ζ in the quadratic factor in the denominator. In this case, the complex conjugated poles move along a semicircle.

$$\zeta = 2(1 - \alpha) \quad Q = \frac{1}{4(1 - \alpha)}$$

```
plot1 := plot([Re(f1(alpha, 68E3, 47E-9)), Im(f1(alpha, 68E3, 47E-9)), alpha=0..1], color="Blue") :
plot2 := plot([Re(f2(alpha, 68E3, 47E-9)), Im(f2(alpha, 68E3, 47E-9)), alpha=0..1], color="Green") :
plots[display](plot1, plot2, thickness=3, font=[Helvetica, roman, 18], axis[2]=[thickness=1.5],
axis[1]=[thickness=1.5], labels=["Re", "Im"], labelfont=["HELVETICA", 18], numpoints
=100, gridlines, size=[600, 300])
```

Problems

Download Maple file from DTU Learn, week 11/exercises.

Adapt the Maple file to plot Bode plot curves for the total transfer function and for the individual factors.

Problem 1

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

$$G(j\omega) = \frac{10^4(j\omega + 2)}{(j\omega + 10)(j\omega + 100)}$$

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

$$H(s) = \frac{100s}{s^2 + 20s + 10000}$$

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

Problem 3 – Sallen-Key Lowpass Filter

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

$$H(s) = \frac{b_0}{s^2 + a_1 s + a_0}$$

$$a_1 = 2827.53$$

$$a_0 = 394815$$

$$b_0 = 789629$$

Problem 4 – Sallen-Key Highpass filter

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

$$H(s) = \frac{2s^2}{\left(s + \frac{\pi}{10}\right)^2}$$

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

Problem 5 – Sallen-Key Bandpass filter

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

$$H(s) = \frac{b_1 s}{s^2 + a_1 s + a_0}$$

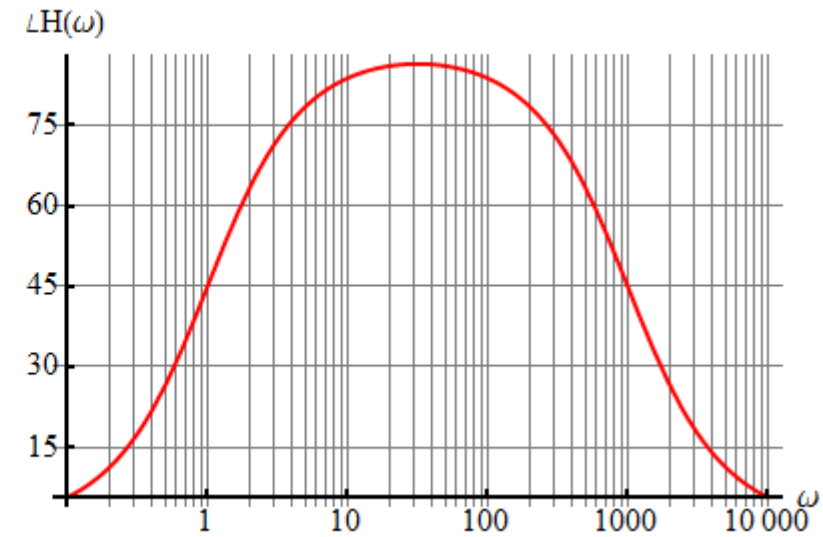
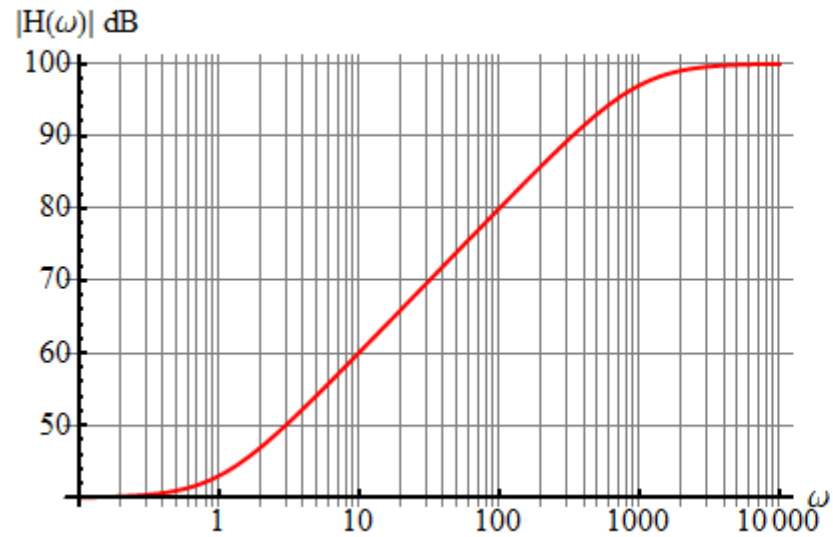
$$a_1 = 31.41$$

$$a_0 = 98695$$

$$b_1 = 628.3$$

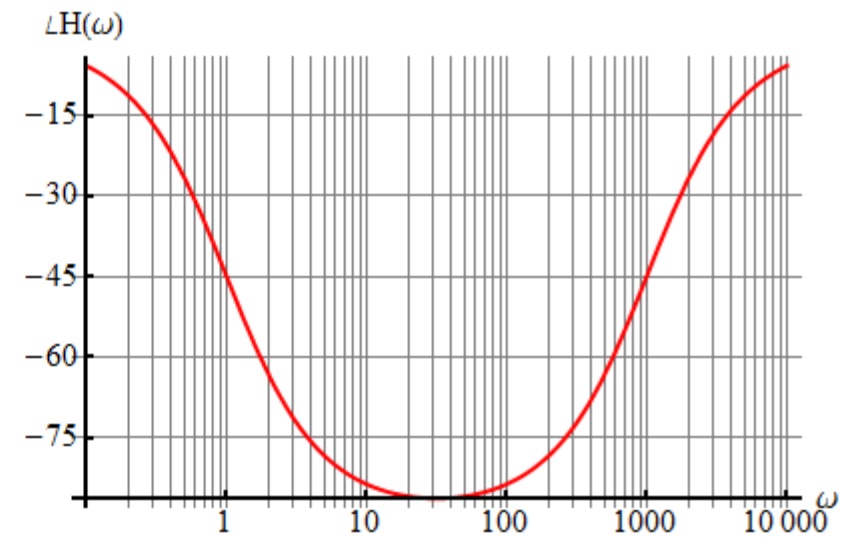
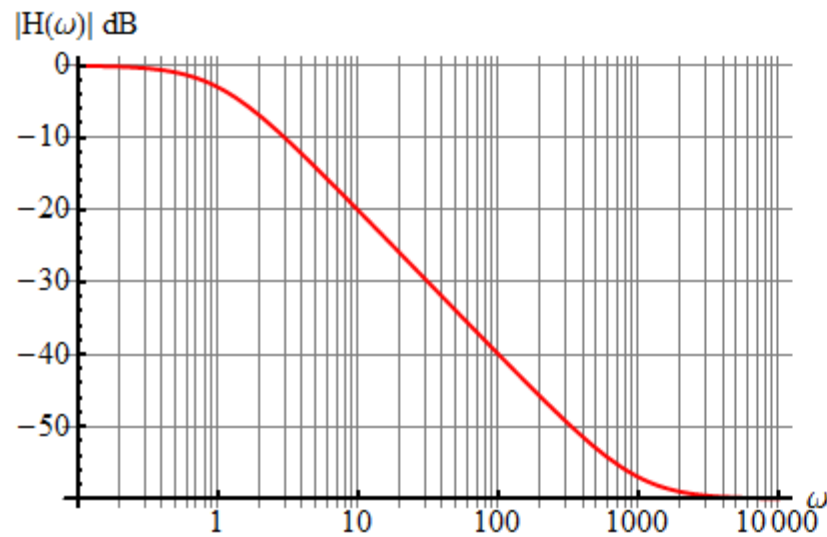
Problem 6 – Systems identification

Use the frequency characteristics to write the transfer function.



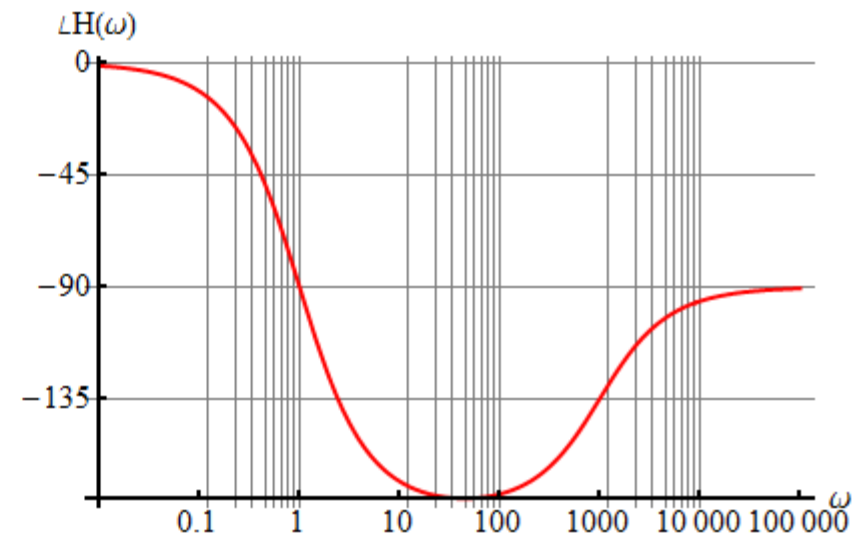
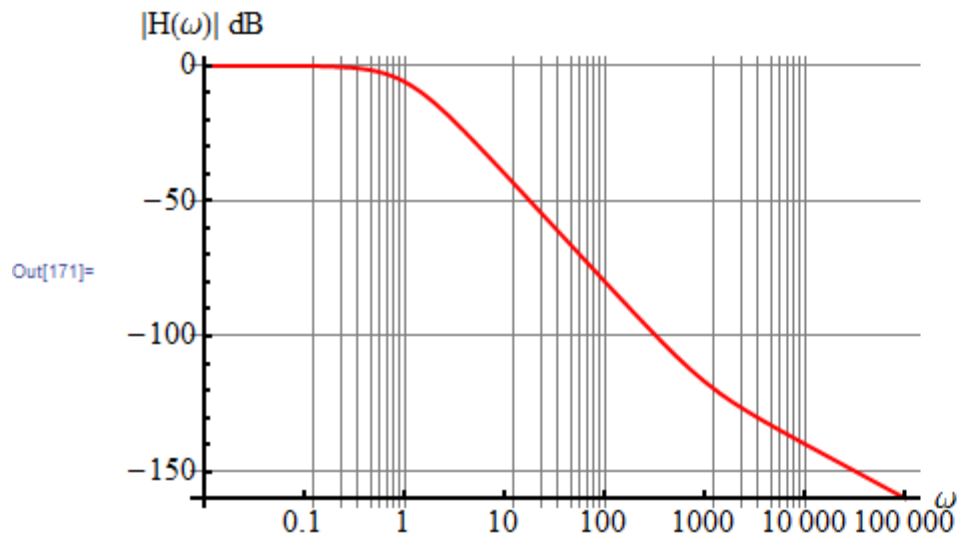
Problem 7 – Systems identification

Use the frequency characteristics to write the transfer function.



Problem 8 – Systems identification

Use the frequency characteristics to write the transfer function.



Solutions

Problem 1

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

$$G(j\omega) = \frac{10^4(j\omega + 2)}{(j\omega + 10)(j\omega + 100)}$$

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

Problem 1 (sol)

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

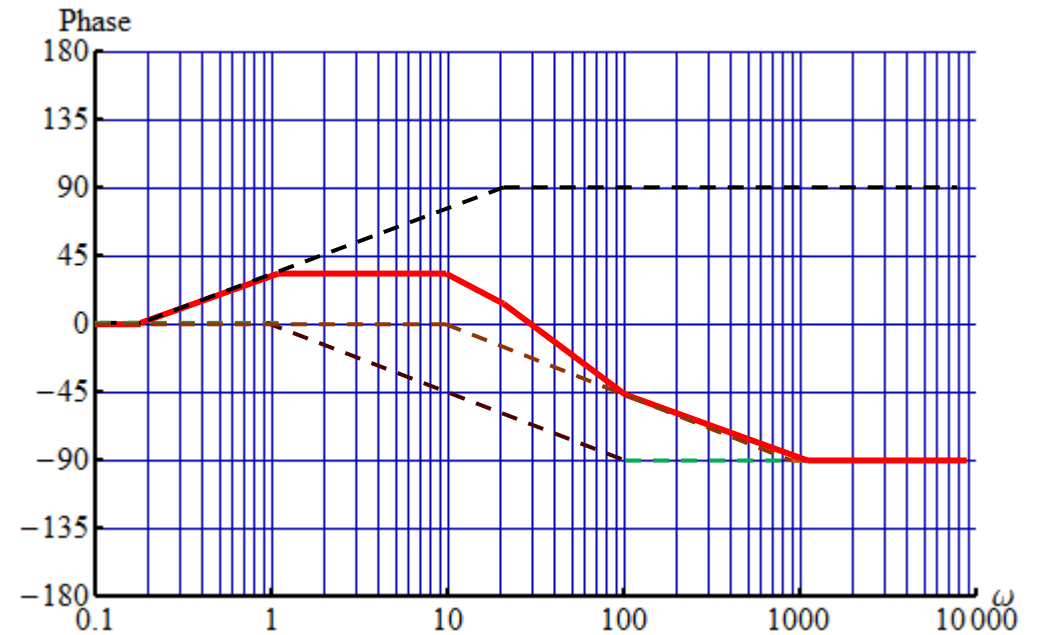
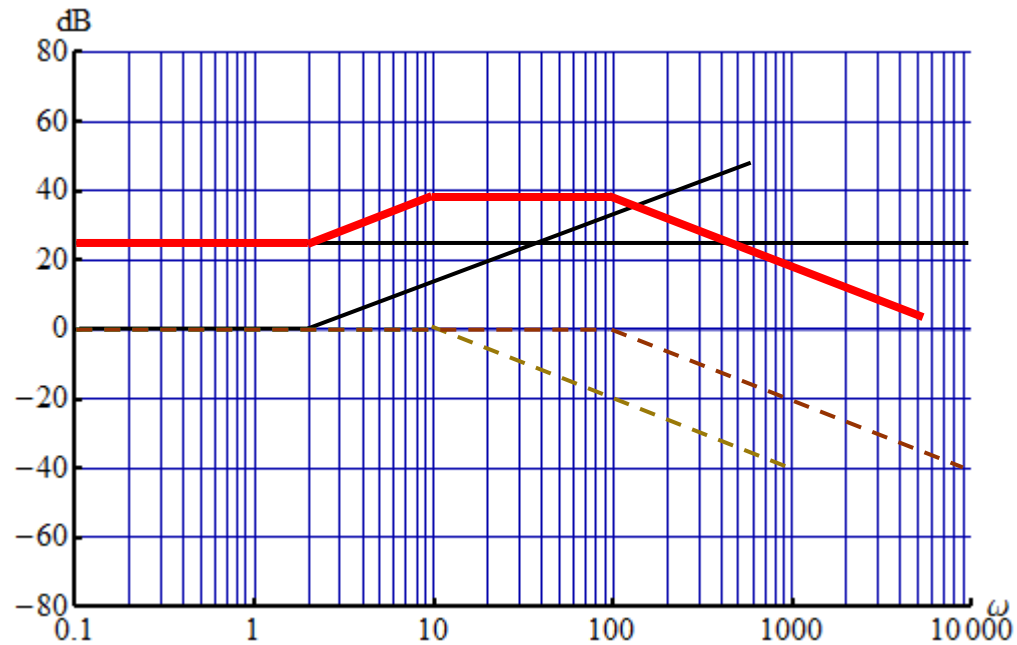
$$G(j\omega) = \frac{10^4(j\omega + 2)}{(j\omega + 10)(j\omega + 100)}$$

$$G(j\omega) = \frac{2 \times 10^4}{10 \times 100} \frac{\left(\frac{j\omega}{2} + 1\right)}{\left(\frac{j\omega}{10} + 1\right)\left(\frac{j\omega}{100} + 1\right)}$$

$$G(j\omega) = 20 \frac{\left(\frac{j\omega}{2} + 1\right)}{\left(\frac{j\omega}{10} + 1\right)\left(\frac{j\omega}{100} + 1\right)}$$

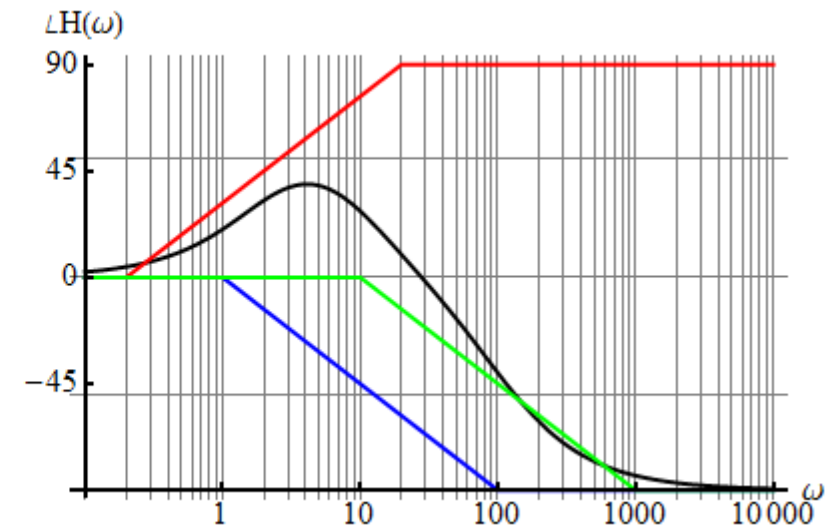
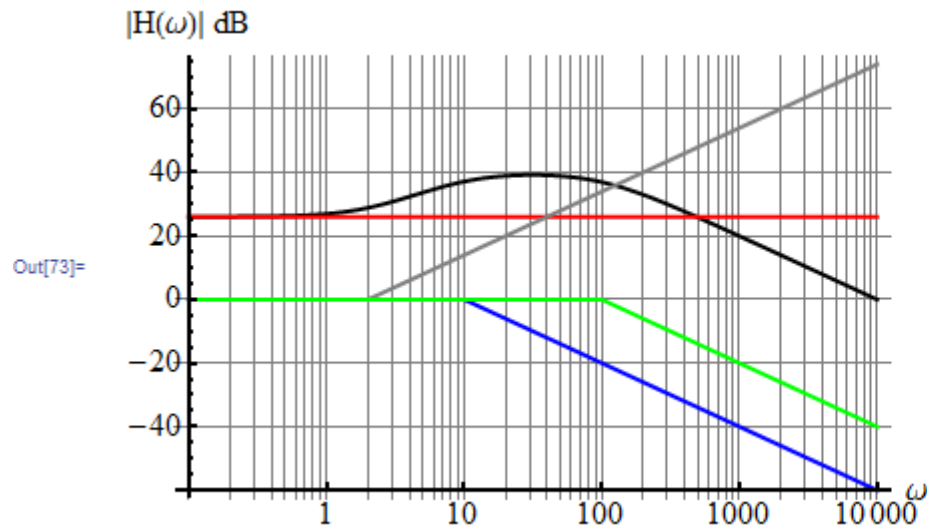
Problem 1 (sol – by hand)

$$G(j\omega) = 20 \frac{\left(\frac{j\omega}{2} + 1\right)}{\left(\frac{j\omega}{10} + 1\right)\left(\frac{j\omega}{100} + 1\right)}$$



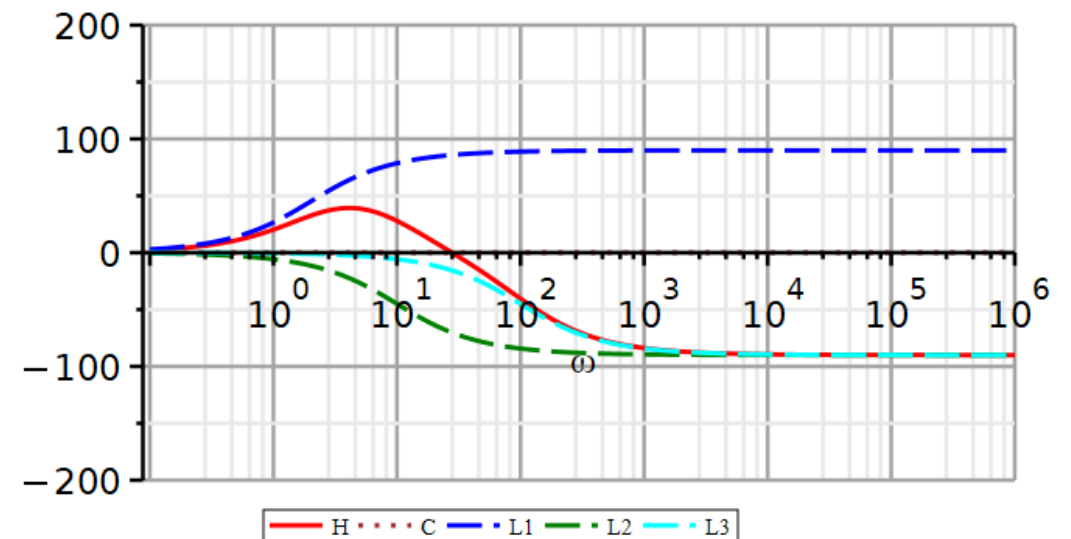
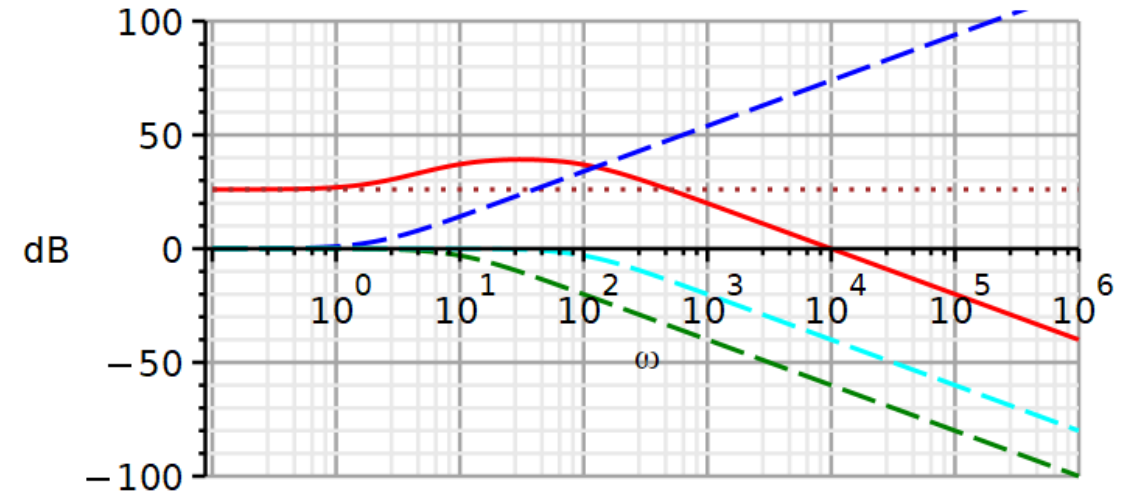
Problem 1 (sol – by Mathematica)

$$G(j\omega) = 20 \frac{\left(\frac{j\omega}{2} + 1\right)}{\left(\frac{j\omega}{10} + 1\right)\left(\frac{j\omega}{100} + 1\right)}$$



Problem 1 (sol by Maple)

$$G(j\omega) = 20 \frac{\left(\frac{j\omega}{2} + 1\right)}{\left(\frac{j\omega}{10} + 1\right)\left(\frac{j\omega}{100} + 1\right)}$$



Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

$$H(s) = \frac{100s}{s^2 + 20s + 10000}$$

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

Problem 2 (sol)

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

$$H(s) = \frac{100s}{s^2 + 20s + 10000}$$

$$H(j\omega) = \frac{100j\omega}{(j\omega)^2 + 20j\omega + 10000}$$

$$H(j\omega) = \frac{\frac{j\omega}{100}}{\left(\frac{j\omega}{100}\right)^2 + \frac{20}{100} \frac{j\omega}{100} + 1}$$

$$H(j\omega) = \frac{\frac{j\omega}{100}}{\left(\frac{j\omega}{100}\right)^2 + 2 \times 0.1 \frac{j\omega}{100} + 1}$$

Damping factor: $\zeta = 0.1$

Undamped resonance frequency: $\omega_n = 100 \text{ rad/s}$

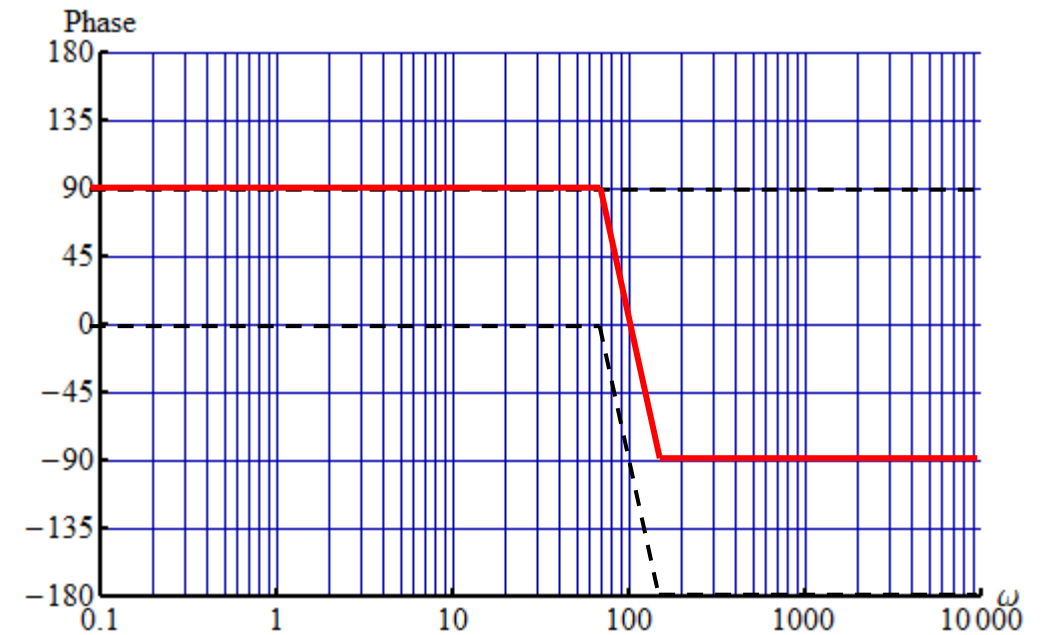
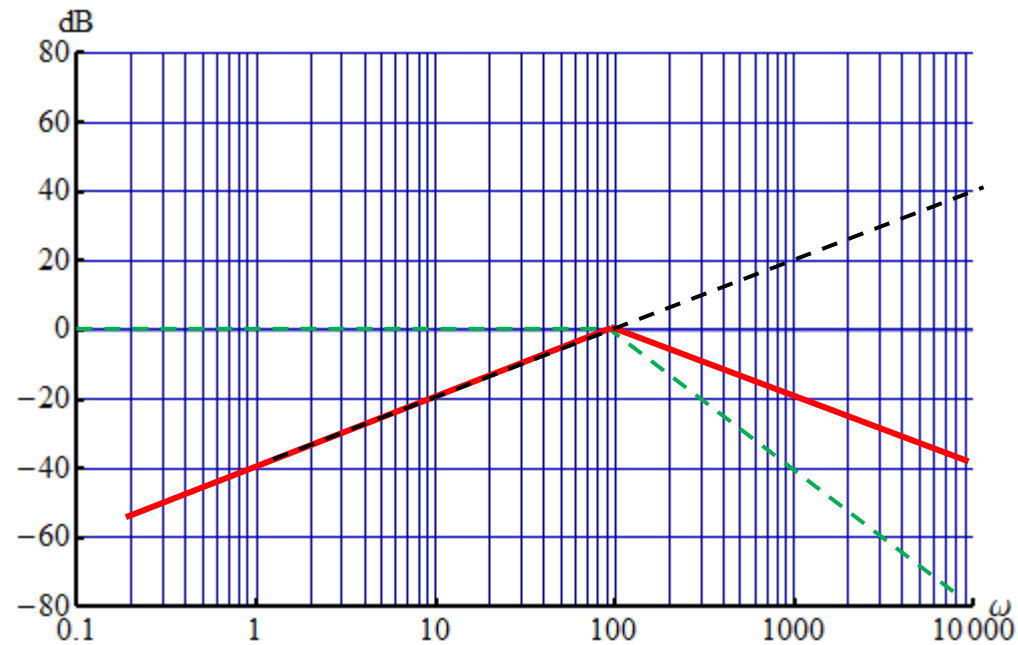
Problem 2 (sol – by hand)

$$H(j\omega) = \frac{\frac{j\omega}{100}}{\left(\frac{j\omega}{100}\right)^2 + 2 \times 0.1 \frac{j\omega}{100} + 1}$$

$$\zeta \leq 0.2: \alpha = 1.410 \zeta - 0.150 \zeta^2 \quad \alpha = 0.1395$$

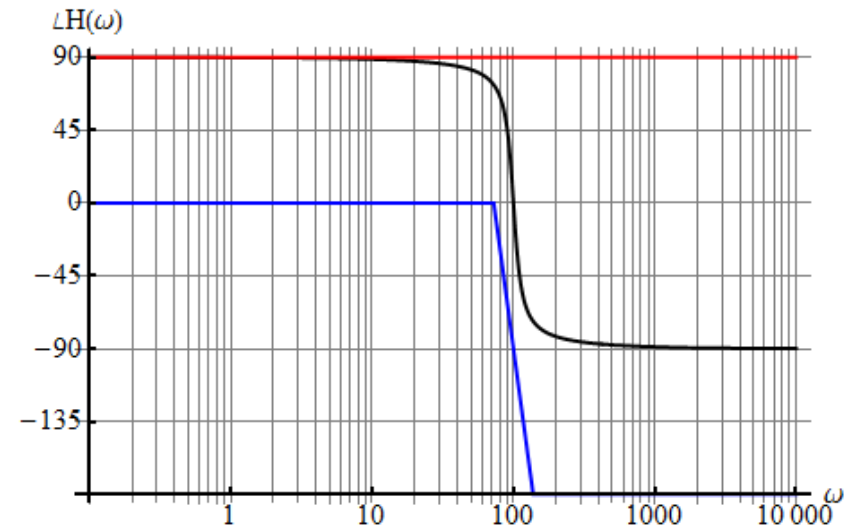
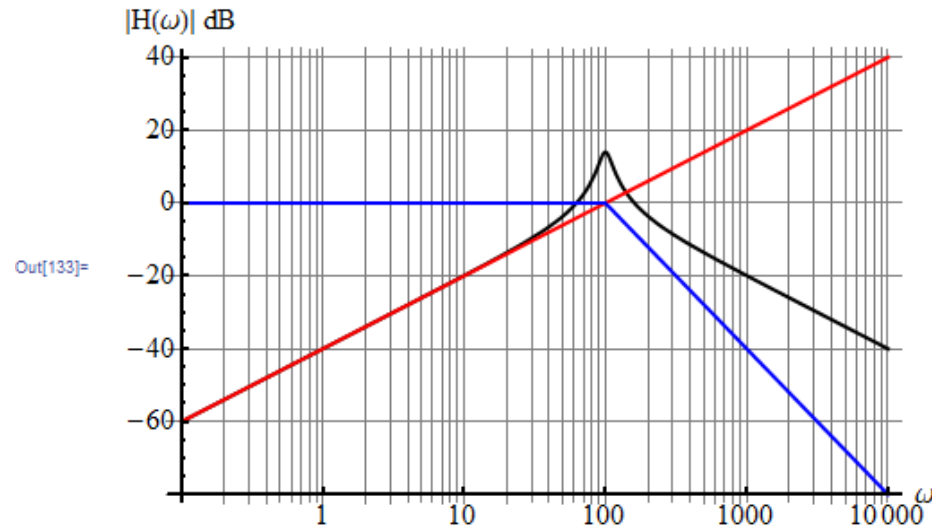
$$\frac{100 \text{ rad/s}}{10^{0.1395}} = 72.5 \text{ rad/s}$$

$$10^{0.1395} 100 \text{ rad/s} = 137.88 \text{ rad/s}$$



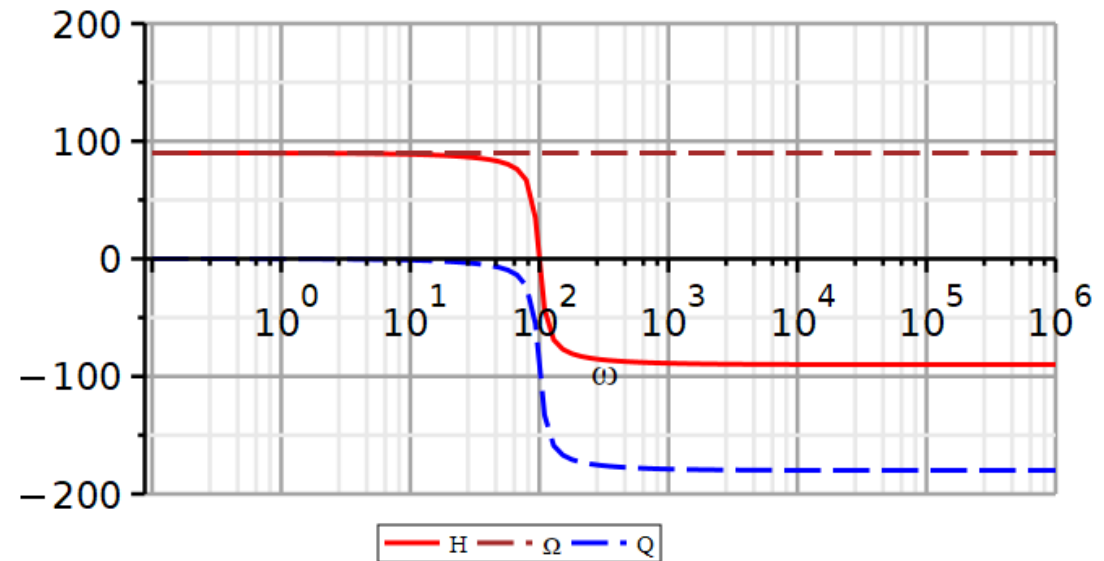
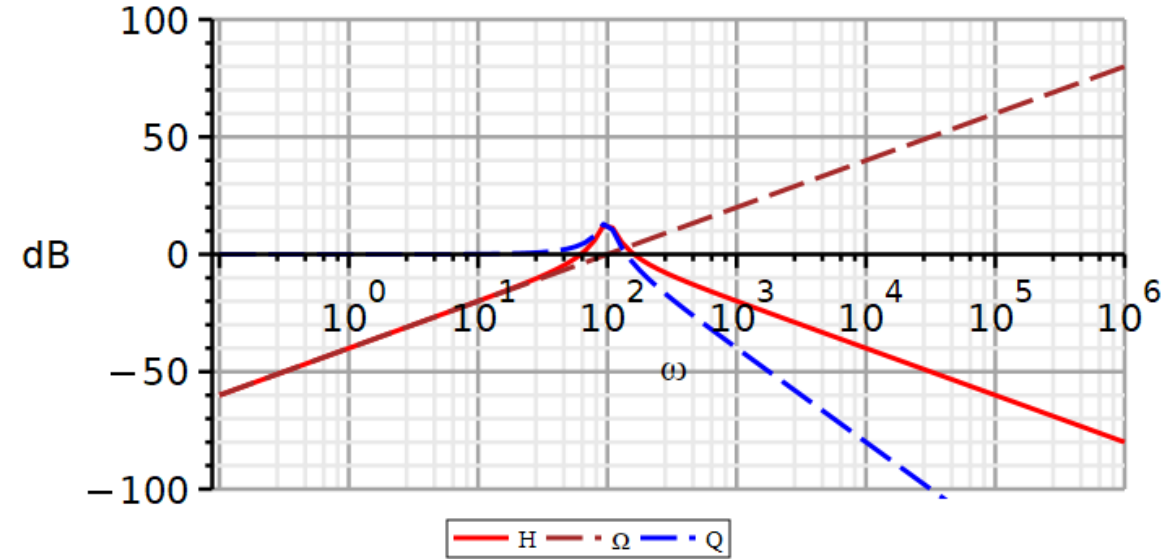
Problem 2 (sol by Mathematica)

$$H(j\omega) = \frac{\frac{j\omega}{100}}{\left(\frac{j\omega}{100}\right)^2 + 2 \times 0.1 \frac{j\omega}{100} + 1}$$



Problem 2 (sol by Maple)

$$H(j\omega) = \frac{\frac{j\omega}{100}}{\left(\frac{j\omega}{100}\right)^2 + 2 \times 0.1 \frac{j\omega}{100} + 1}$$



Problem 3 – Sallen-Key Lowpass Filter

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

$$H(s) = \frac{b_0}{s^2 + a_1s + a_0}$$

$$a_1 = 2827.53$$

$$a_0 = 394815$$

$$b_0 = 789629$$

Problem 3 – Sallen-Key Lowpass Filter (sol)

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

$$H(s) = \frac{b_0}{s^2 + a_1s + a_0}$$

$$H(s) = \frac{789629}{(s + 147.3)(s + 2680)}$$

$$H(s) = \frac{2}{\left(\frac{s}{147.3} + 1\right)\left(\frac{s}{2680} + 1\right)}$$

$$H(j\omega) = \frac{2}{\left(\frac{j\omega}{147.3} + 1\right)\left(\frac{j\omega}{2680} + 1\right)}$$

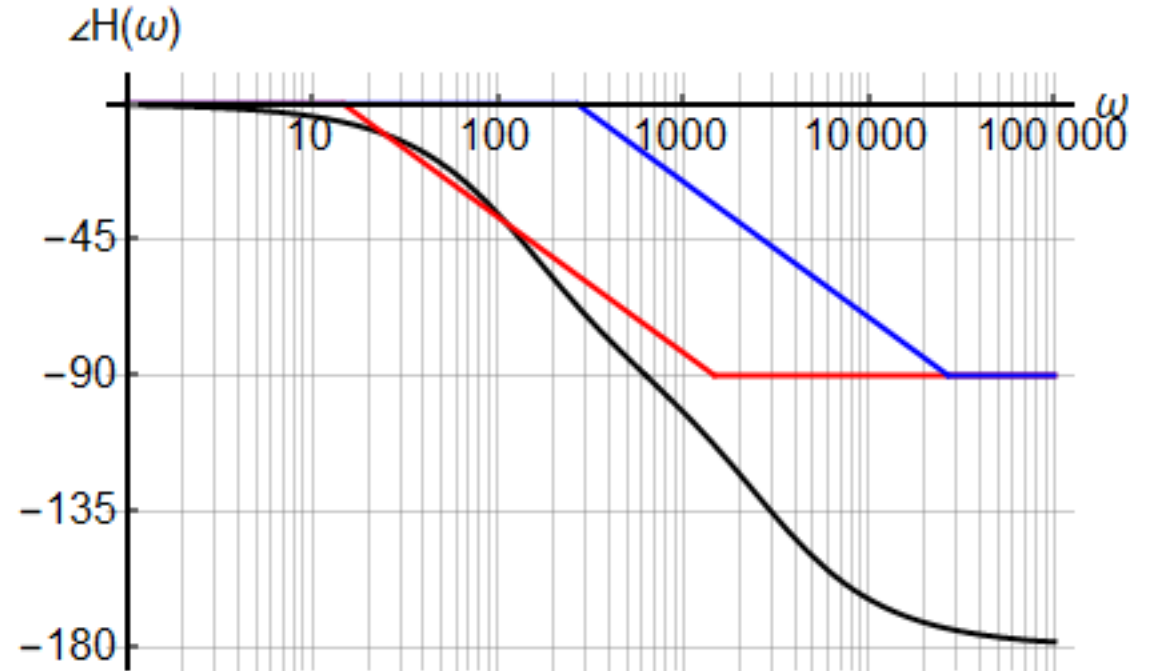
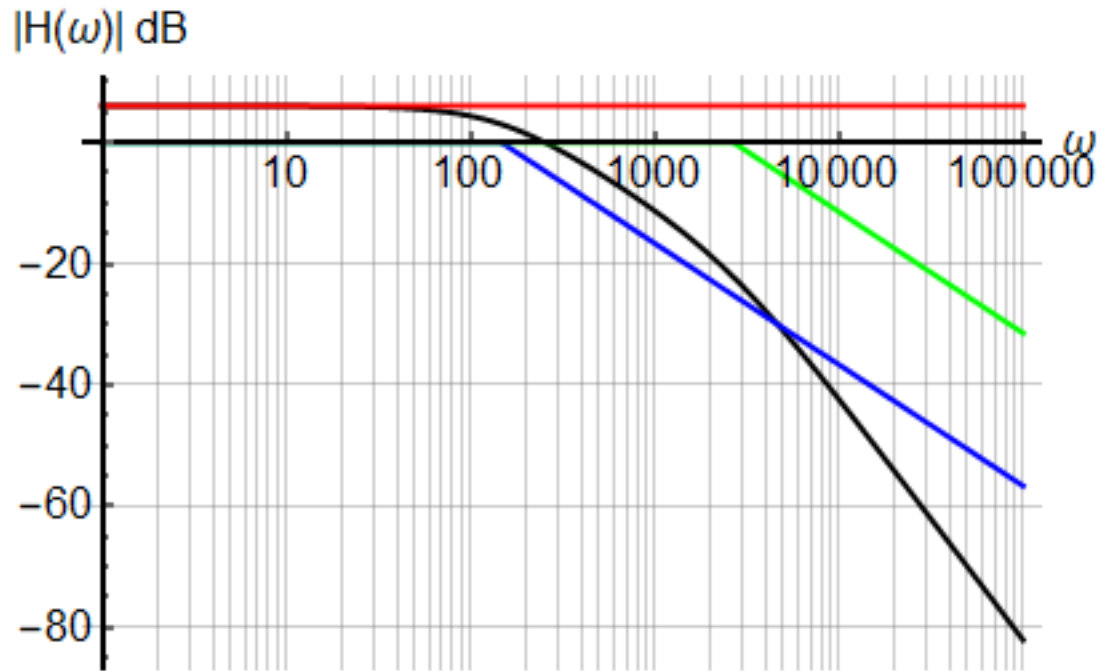
$$a_1 = 2827.53$$

$$a_0 = 394815$$

$$b_0 = 789629$$

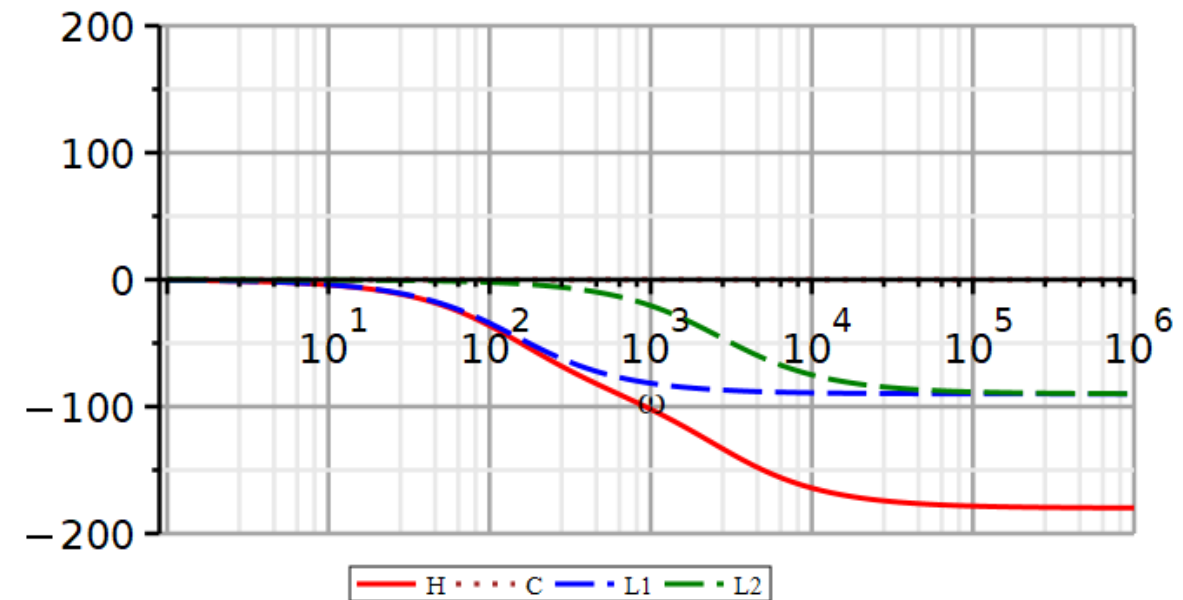
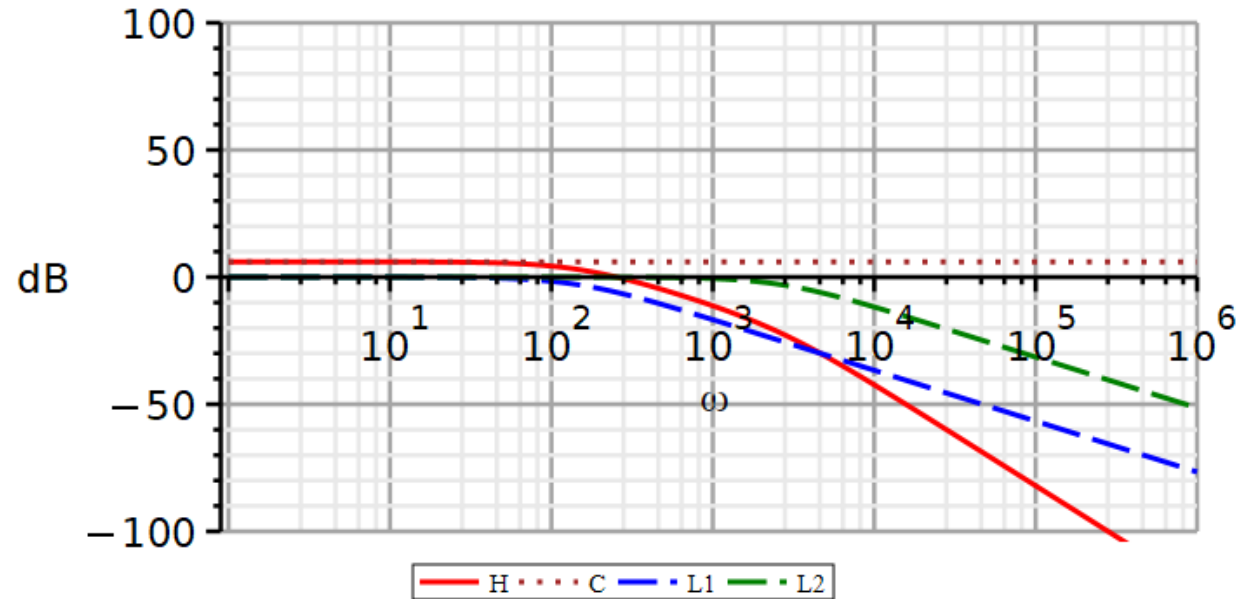
Problem 3 – Sallen-Key Lowpass Filter (sol – by Mathematica)

$$H(j\omega) = \frac{2}{\left(\frac{j\omega}{147.3} + 1\right)\left(\frac{j\omega}{2680} + 1\right)}$$



Problem 3 – Sallen-Key Lowpass Filter (sol – by Maple)

$$H(j\omega) = \frac{2}{\left(\frac{j\omega}{147.3} + 1\right)\left(\frac{j\omega}{2680} + 1\right)}$$



Problem 4 – Sallen-Key Highpass filter

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

$$H(s) = \frac{2s^2}{\left(s + \frac{\pi}{10}\right)^2}$$

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

Problem 4 – Sallen-Key Highpass filter (sol)

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

$$H(s) = \frac{2s^2}{\left(s + \frac{\pi}{10}\right)^2}$$

$$H(s) = \frac{2}{\left(\frac{\pi}{10}\right)^2} \frac{s^2}{\left(\frac{s}{\frac{\pi}{10}} + 1\right)^2}$$

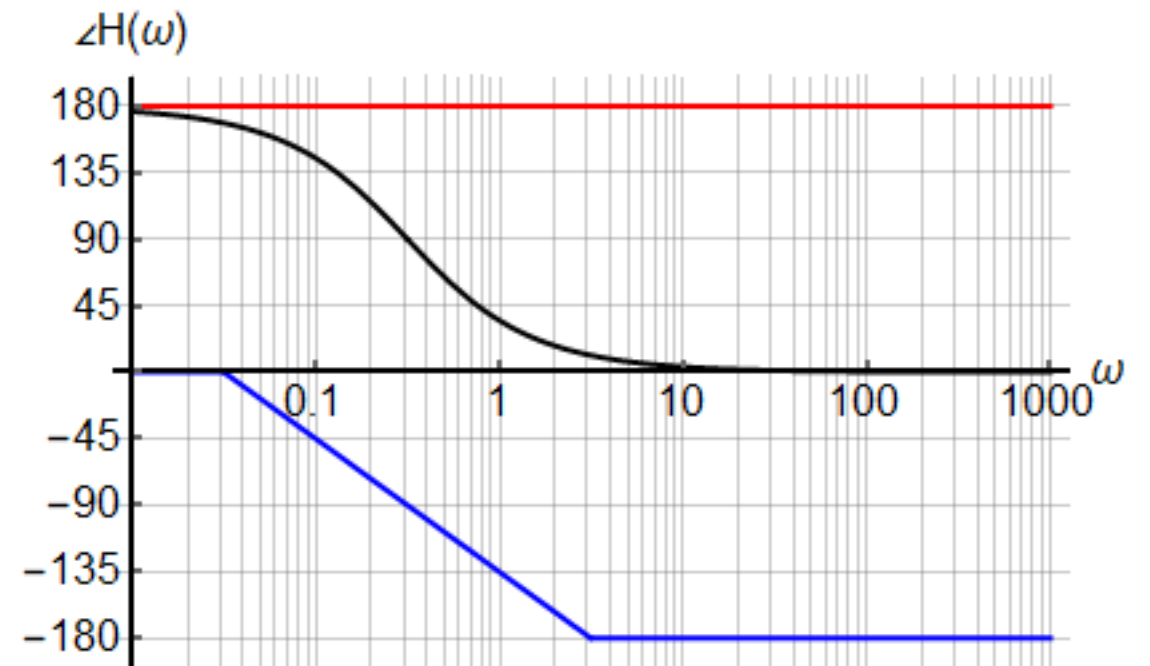
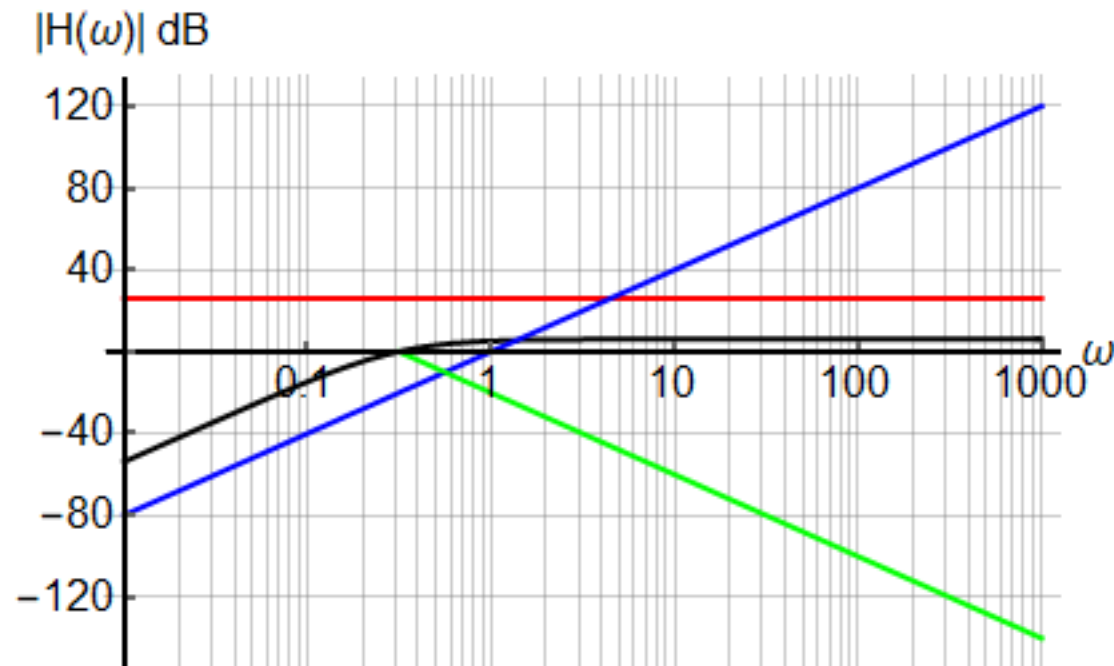
$$H(j\omega) = \frac{2}{\left(\frac{\pi}{10}\right)^2} \frac{(j\omega)^2}{\left(\frac{j\omega}{\frac{\pi}{10}} + 1\right)^2}$$

Alternatively:

$$H(j\omega) = 2 \frac{\left(\frac{j\omega}{10}\right)^2}{\left(\frac{j\omega}{\pi} + 1\right)^2}$$

Problem 4 – Sallen-Key Highpass filter (sol by Mathematica)

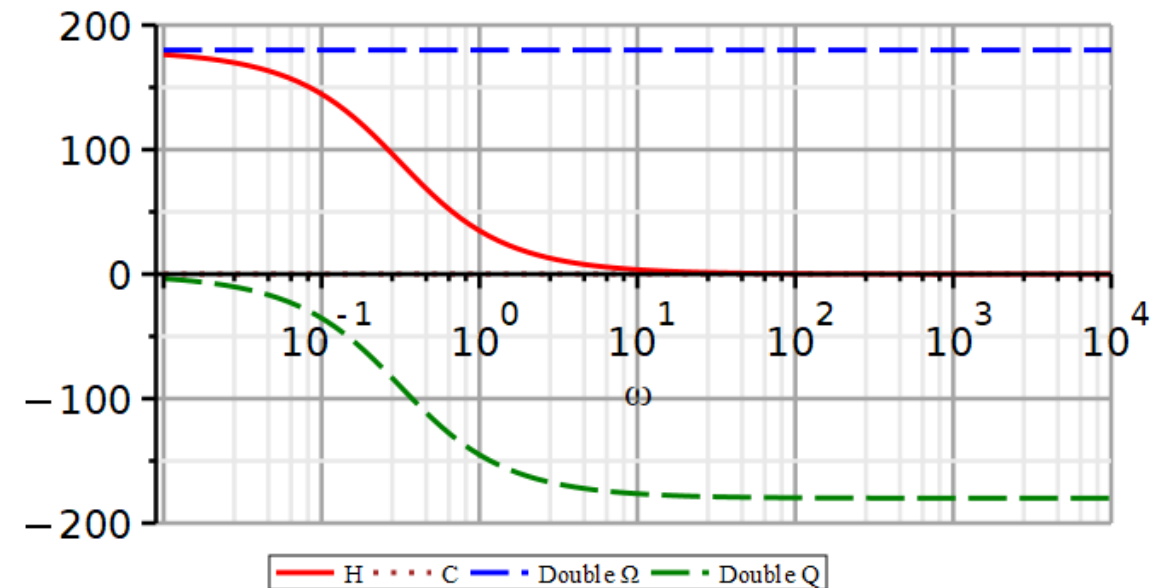
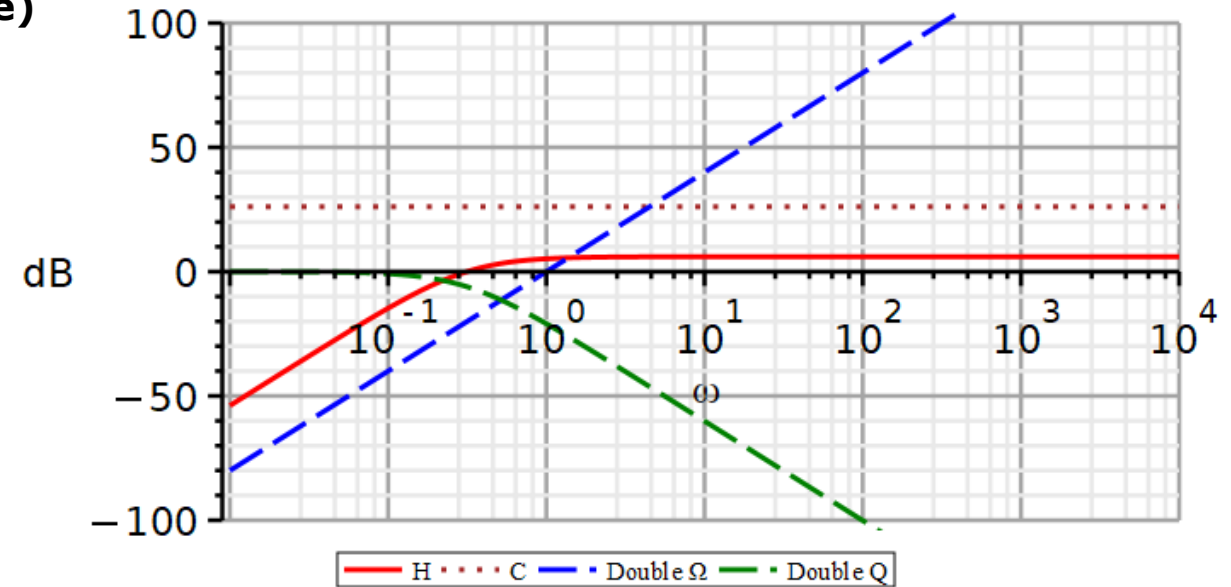
$$H(j\omega) = \frac{2}{\left(\frac{\pi}{10}\right)^2} \frac{(j\omega)^2}{\left(\frac{j\omega}{10} + 1\right)^2} \quad \frac{2}{\left(\frac{\pi}{10}\right)^2} = 20.26$$



Problem 4 – Sallen-Key Highpass filter (sol by Maple)

$$H(j\omega) = \frac{2}{\left(\frac{\pi}{10}\right)^2} \frac{(j\omega)^2}{\left(\frac{j\omega}{\frac{\pi}{10}} + 1\right)^2}$$

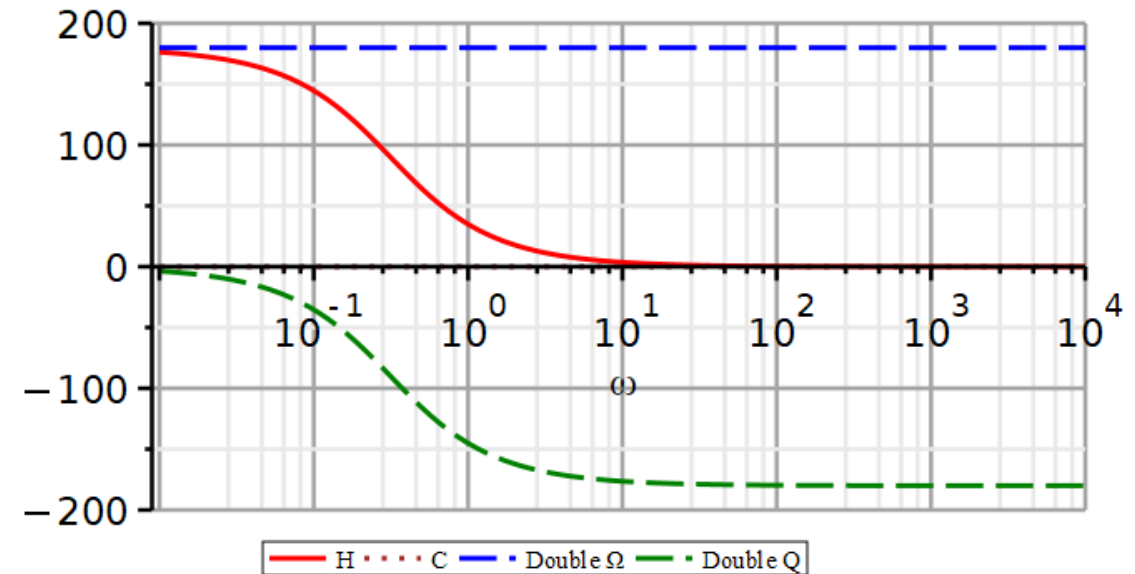
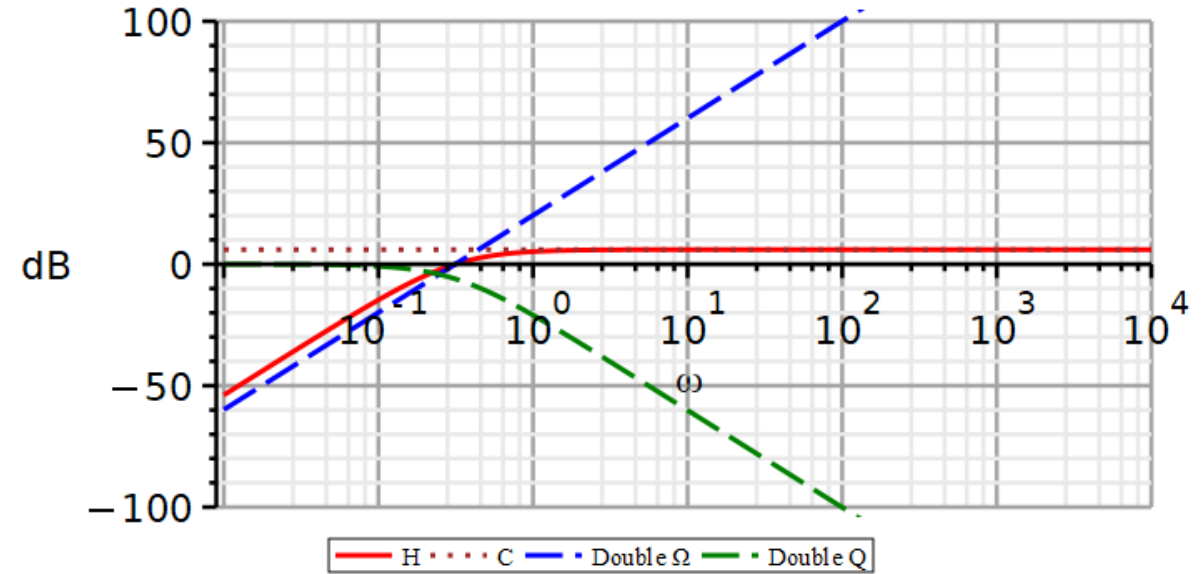
$$\frac{2}{\left(\frac{\pi}{10}\right)^2} = 20.26 \approx 26\text{dB}$$



Problem 4 – Sallen-Key Highpass filter (sol by Maple)

An alternative form:

$$H(j\omega) = 2 \frac{\left(\frac{j\omega}{\pi}\right)^2}{\left(\frac{j\omega}{\pi} + 1\right)^2}$$



Problem 5 – Sallen-Key Bandpass filter

Use the downloaded Maple file to plot the amplitude and phase characteristics of the following transfer function as well as its individual factors.

Start by converting the transfer function to standard Bode format.

In the Maple file:

- Edit the functions to be plotted
- Edit the legends to fit the number of curves
- Edit the line styles to fit the number of curves
- Edit the plotting statements to fit the number of curves
- Edit the frequency range if needed

$$H(s) = \frac{b_1 s}{s^2 + a_1 s + a_0}$$

$$a_1 = 31.41$$

$$a_0 = 98695$$

$$b_1 = 628.3$$

Problem 5 – Sallen-Key Bandpass filter (sol)

$$a_1 = 31.41$$

$$a_0 = 98695$$

$$b_1 = 628.3$$

$$\omega_n = 314.157$$

$$\zeta = 0.05$$

$$\omega_L = 10^{-a} \omega_n \wedge \omega_H = 10^a \omega_n$$

$$a = \begin{cases} 1.410\zeta - 0.150\zeta^2 & \zeta \leq 0.2 \\ 1.475\zeta - 0.475\zeta^2 & \zeta > 0.2 \end{cases}$$

$$a = 0.0701 \Rightarrow \omega_L = 267.3, \omega_H = 369.2$$

$$H(s) = \frac{b_1 s}{s^2 + a_1 s + a_0}$$

$$H(s) = \frac{b_1 s}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

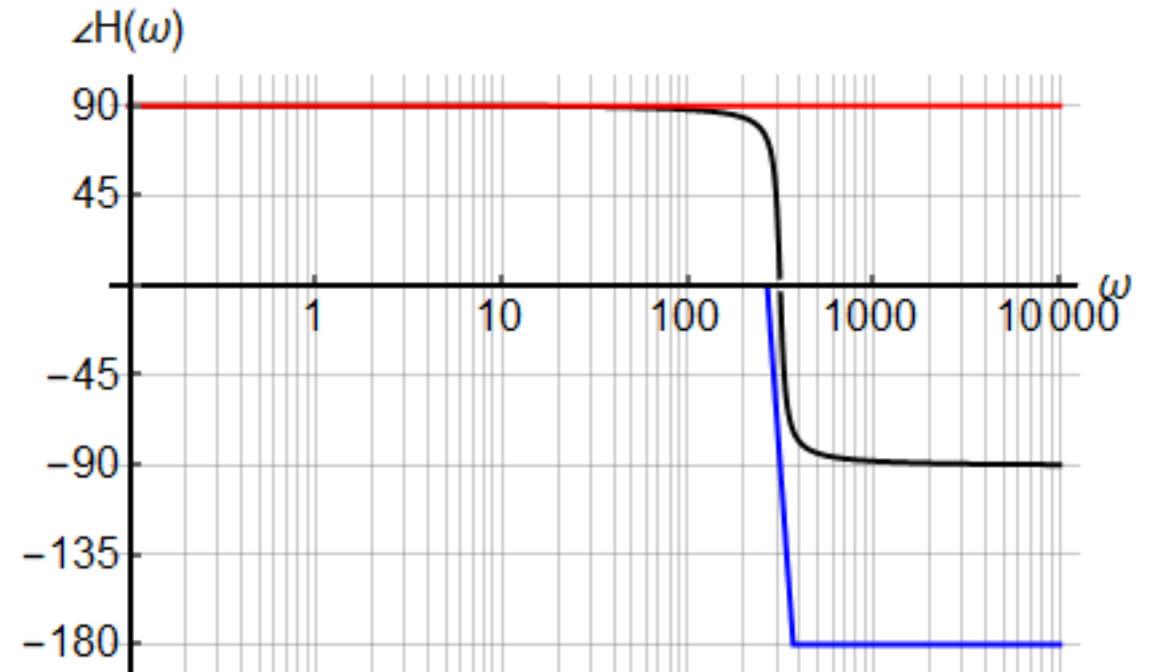
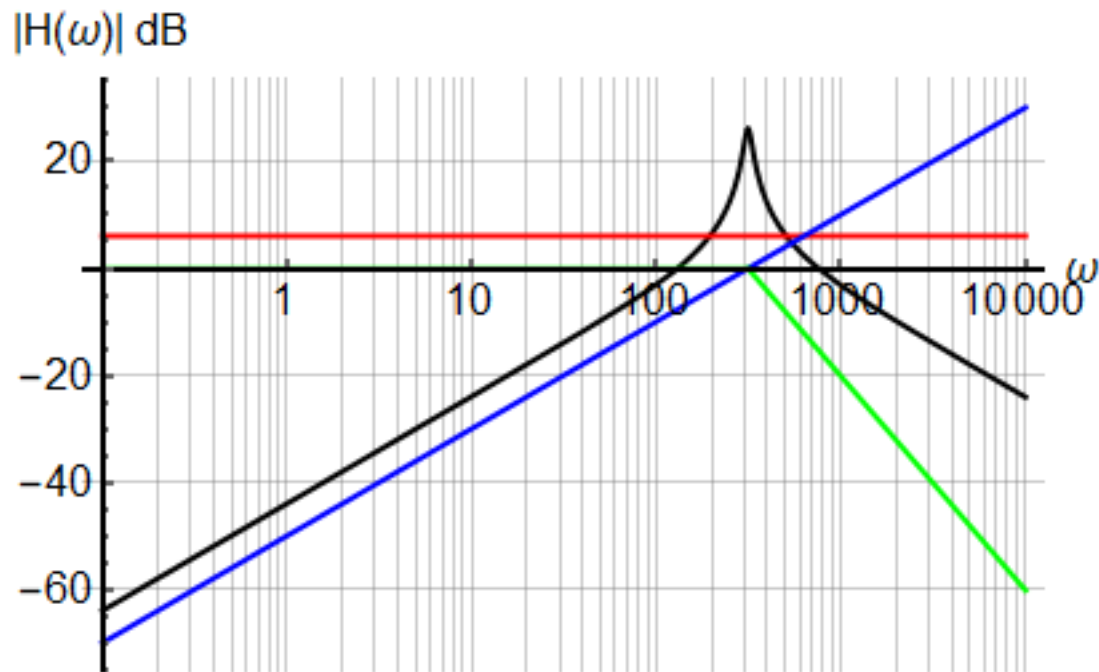
$$H(s) = \frac{b_1}{\omega_n^2} \frac{s}{\left(\frac{s}{\omega_n}\right)^2 + 2\zeta \frac{s}{\omega_n} + 1}$$

$$H(s) = \frac{b_1}{\omega_n} \frac{\frac{s}{\omega_n}}{\left(\frac{s}{\omega_n}\right)^2 + 2\zeta \frac{s}{\omega_n} + 1}$$

$$H(j\omega) = \frac{b_1}{\omega_n} \frac{\frac{j\omega}{\omega_n}}{\left(\frac{j\omega}{\omega_n}\right)^2 + 2\zeta \frac{j\omega}{\omega_n} + 1}$$

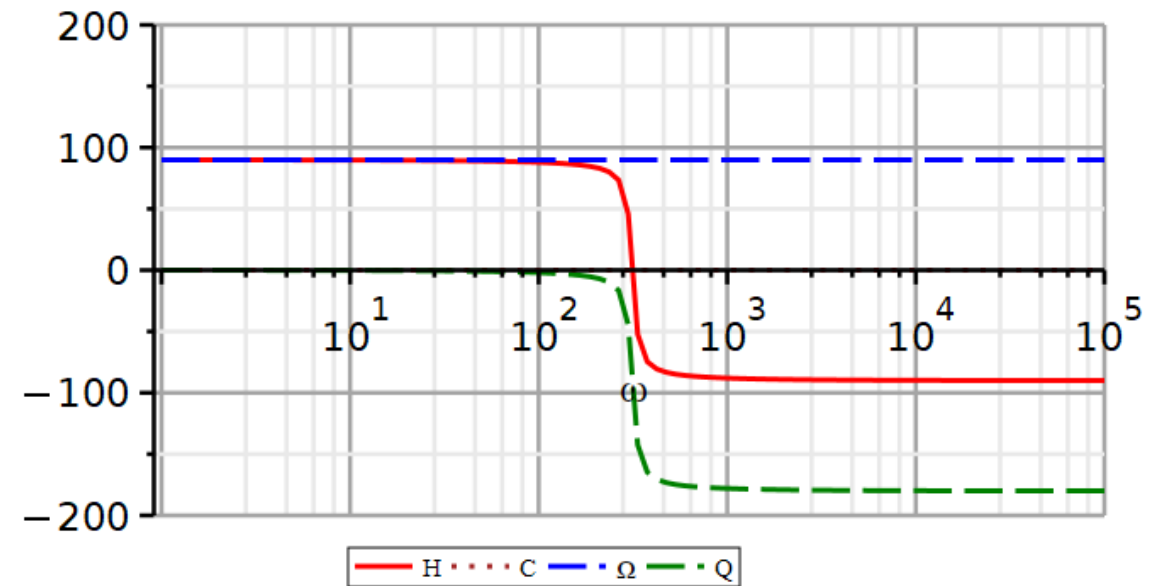
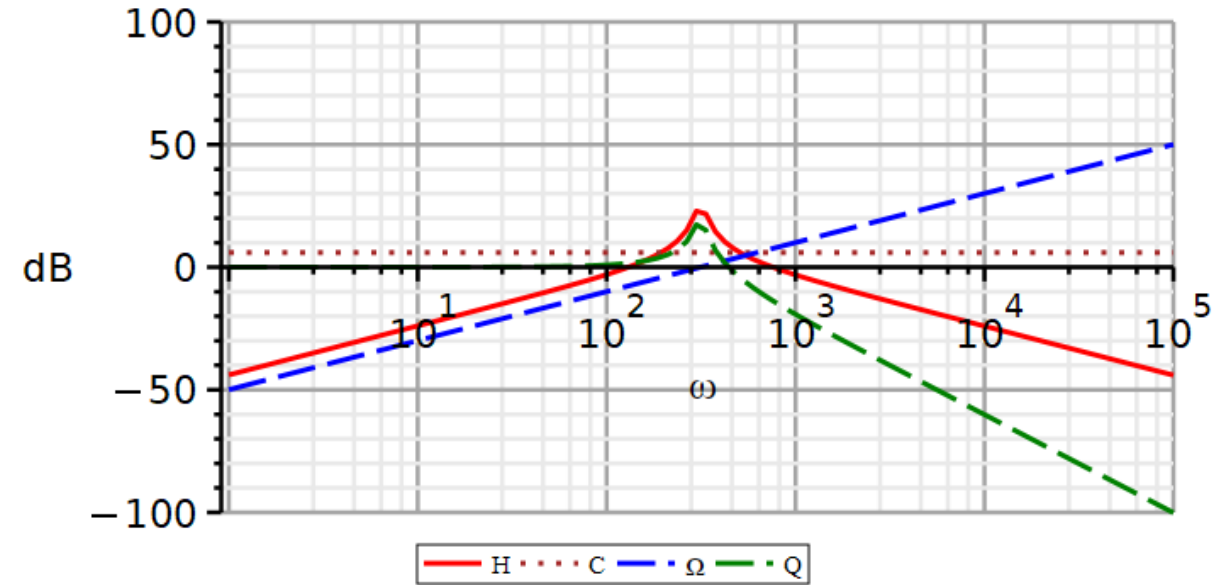
Problem 5 – Sallen-Key Bandpass filter (sol by Mathematica)

$$H(j\omega) = \frac{628}{314} \cdot \frac{\frac{j\omega}{314}}{\left(\frac{j\omega}{314}\right)^2 + 2\zeta \frac{j\omega}{314} + 1} = 2 \frac{\frac{j\omega}{314}}{\left(\frac{j\omega}{314}\right)^2 + 2\zeta \frac{j\omega}{314} + 1}$$



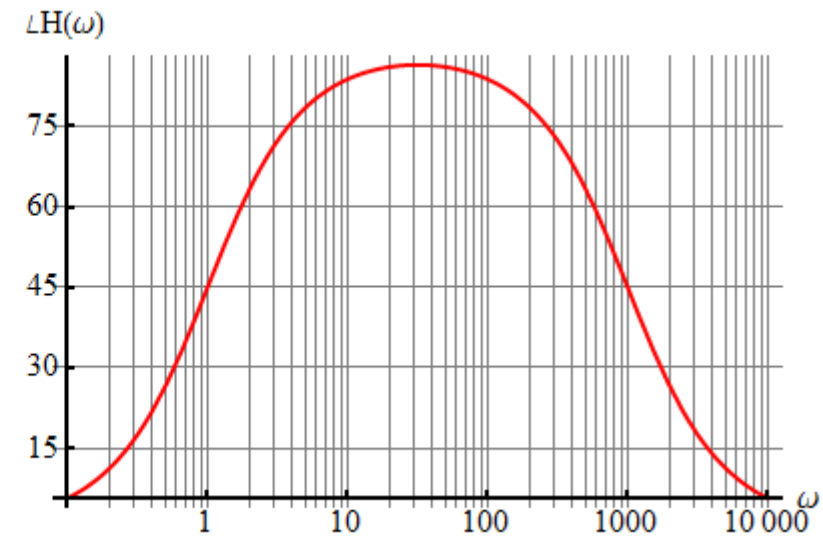
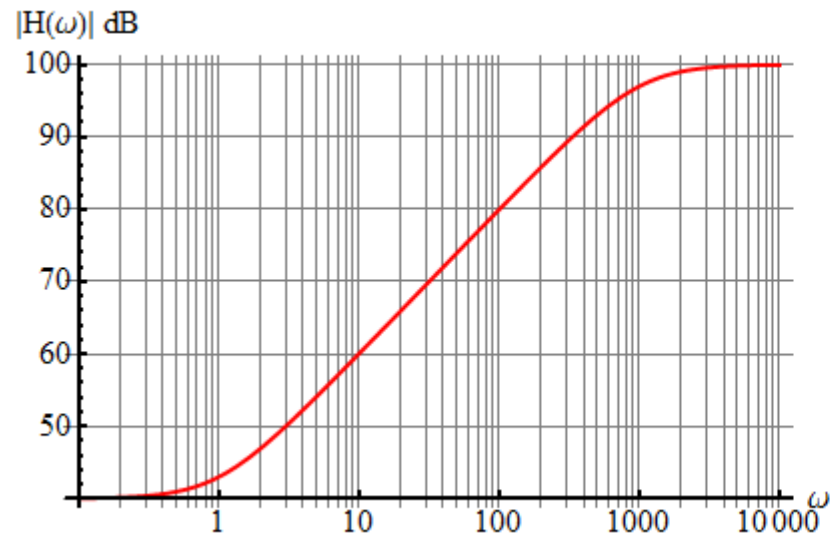
Problem 5 – Sallen-Key Bandpass filter (sol by Maple)

$$H(j\omega) = 2 \frac{\frac{j\omega}{314}}{\left(\frac{j\omega}{314}\right)^2 + 2\zeta \frac{j\omega}{314} + 1}$$



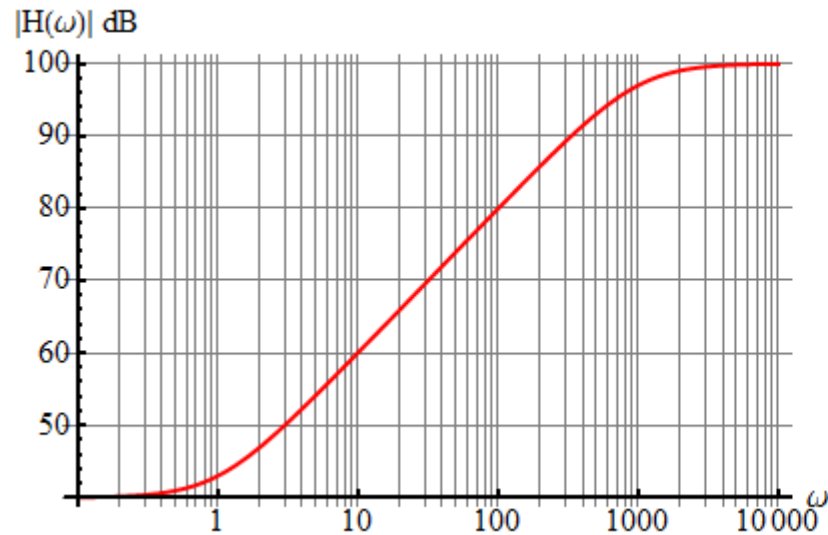
Problem 6

Use the frequency characteristics to write the transfer function.

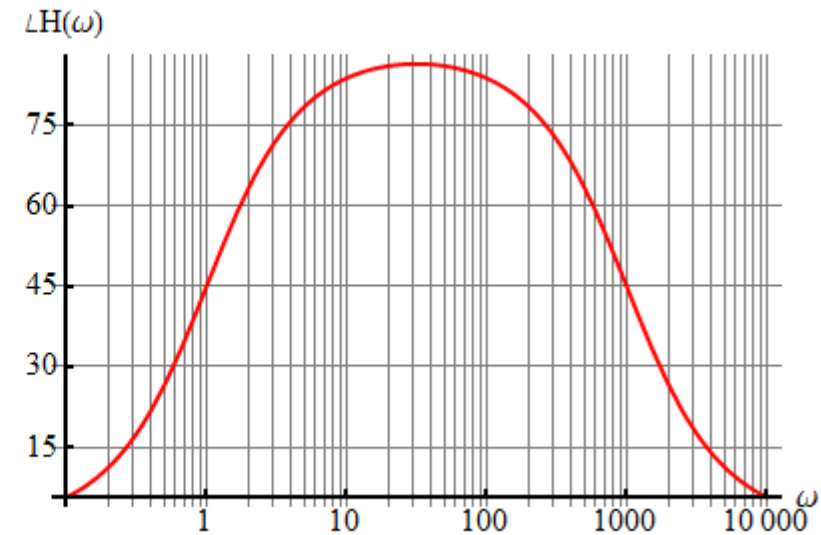


Problem 6 (sol)

Use the frequency characteristics to write the transfer function.



$$H(\omega) = 100 \frac{\frac{j\omega}{1} + 1}{\frac{j\omega}{1000} + 1}$$

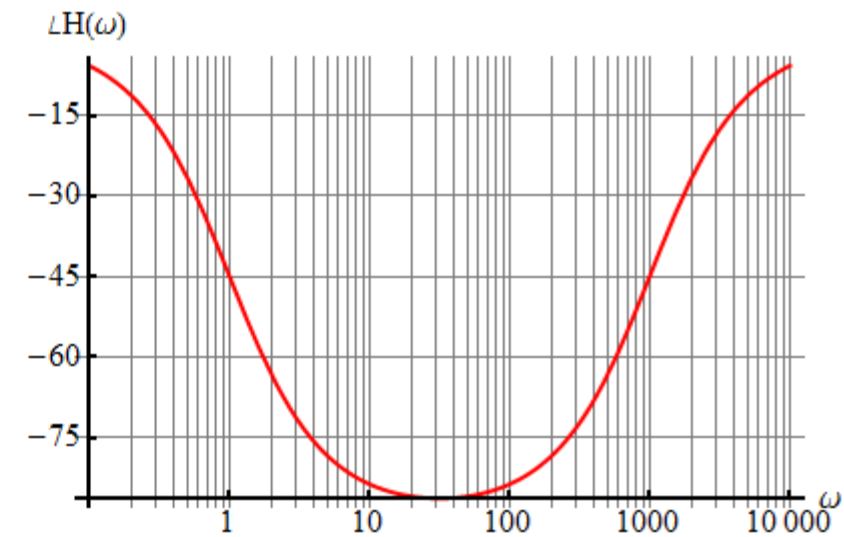
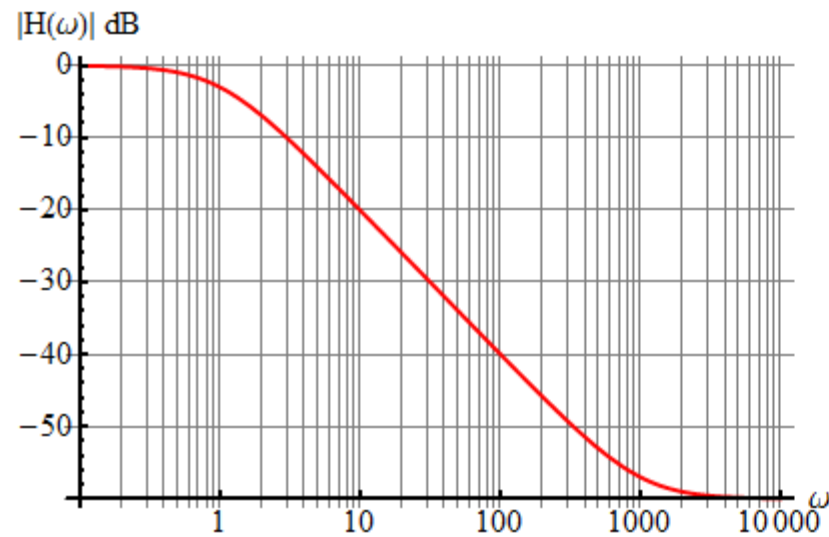


$$H(s) = 10^5 \frac{s + 1}{s + 1000}$$

You should understand on what grounds this solution is constructed.

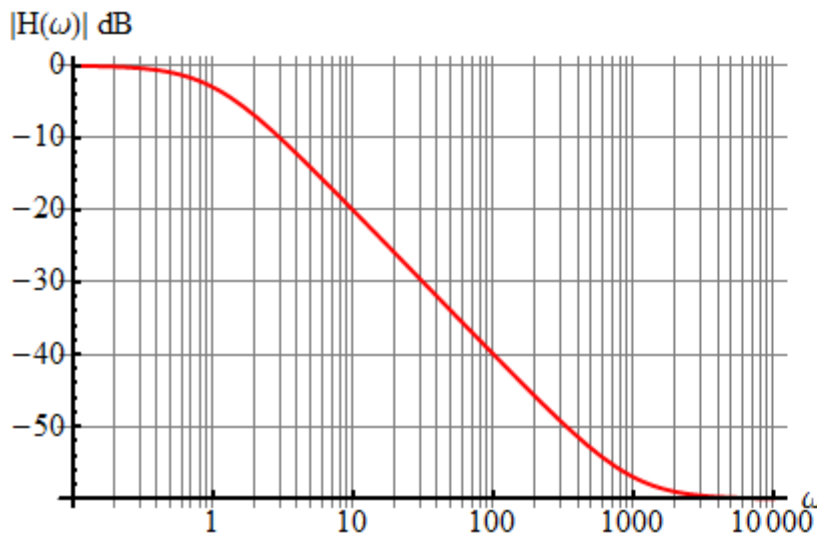
Problem 7

Use the frequency characteristics to write the transfer function.

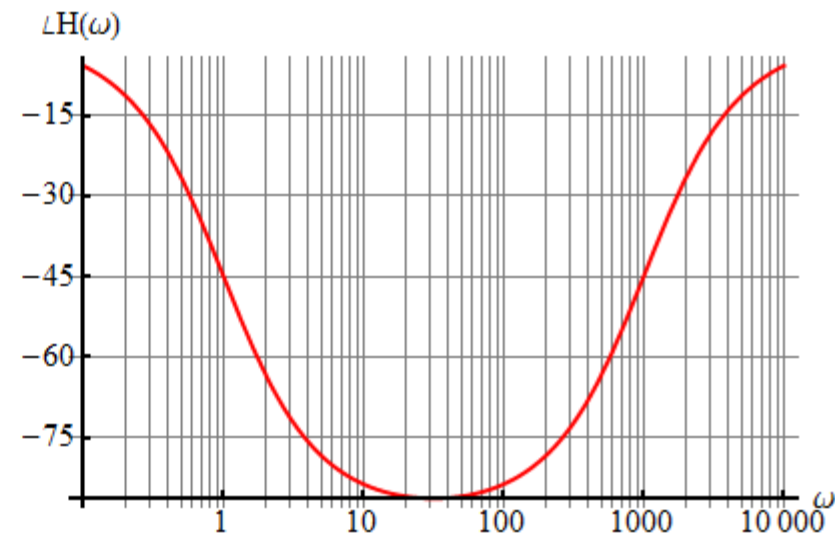


Problem 7 (sol)

Use the frequency characteristics to write the transfer function.



$$H(\omega) = \frac{\frac{j\omega}{1000} + 1}{\frac{j\omega}{1} + 1}$$

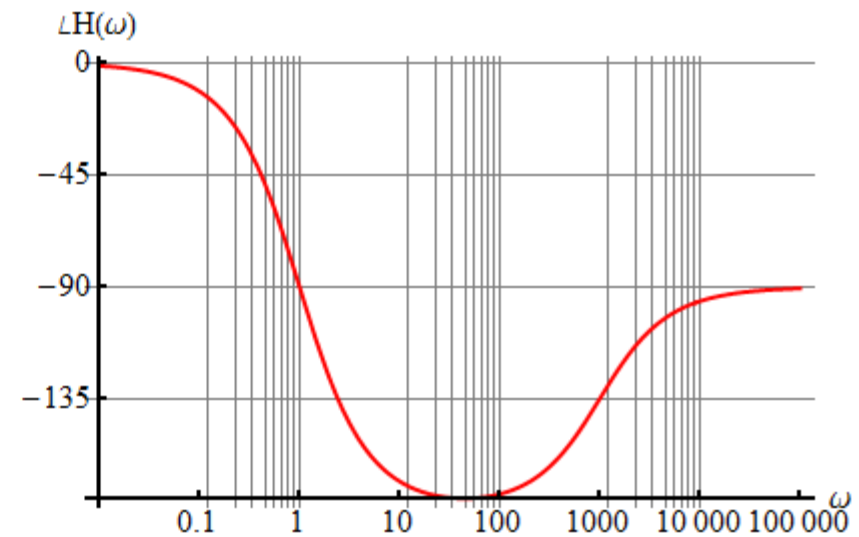
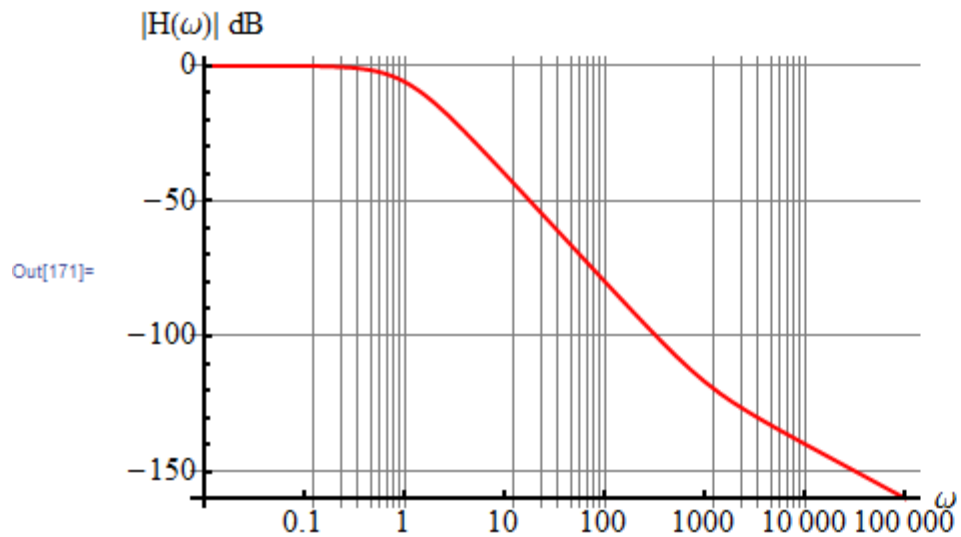


$$H(s) = 10^{-3} \frac{s + 1000}{s + 1}$$

You should understand on what grounds this solution is constructed.

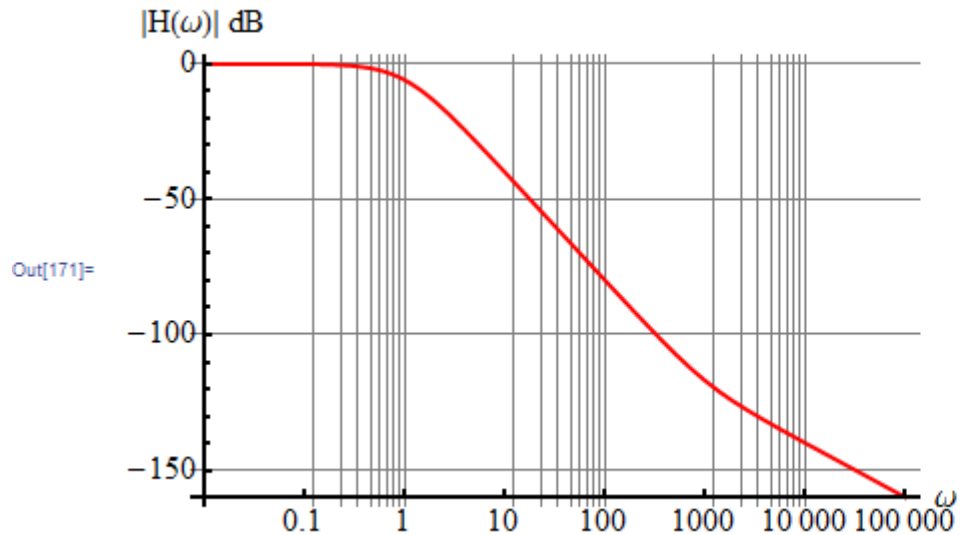
Problem 8

Use the frequency characteristics to write the transfer function.

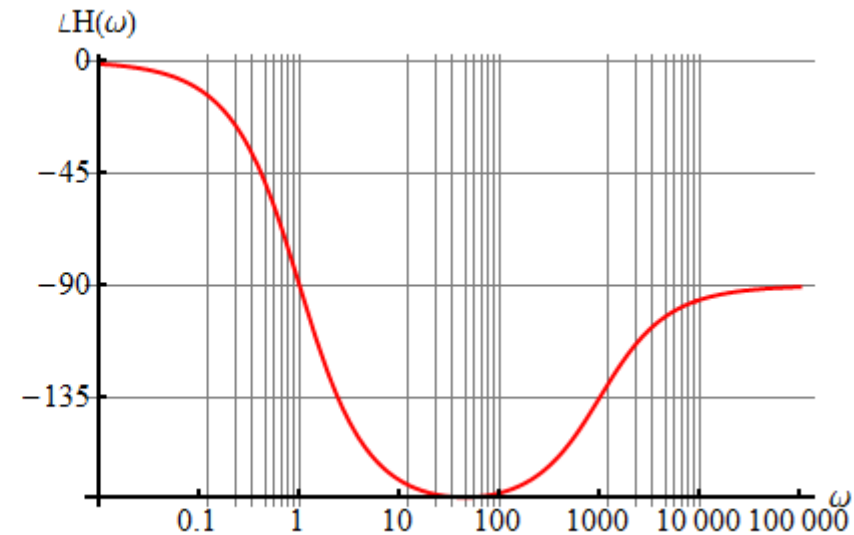


Problem 8 (sol)

Use the frequency characteristics to write the transfer function.



$$H(\omega) = \frac{\frac{j\omega}{1000} + 1}{\left(\frac{j\omega}{1} + 1\right)^2}$$



$$H(s) = 10^{-3} \frac{s + 1000}{(s + 1)^2}$$

The phase changes over two decades, suggesting it could be double pole. But it is only an estimate. It could also be a quadratic factor with a damping factor ζ very close to but slightly less than 1.